

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: DATA VISUALIZATION WITH POWER BI & TABLEAU

Practical No. 4

Aim: Performing Data Analysis using Calculations in Tableau

- a) Create basic calculated fields using formulas
- b) Perform row-level calculations
- c) Perform aggregate-level calculations
- d) Implement FIXED Level of Detail (LOD) calculations
- e) Use INCLUDE and EXCLUDE LOD expressions
- f) Create and use parameters for dynamic analysis
- g) Apply calculations to fix data issues and enhance insights

Dataset: fast_food_ordering_dataset.csv.

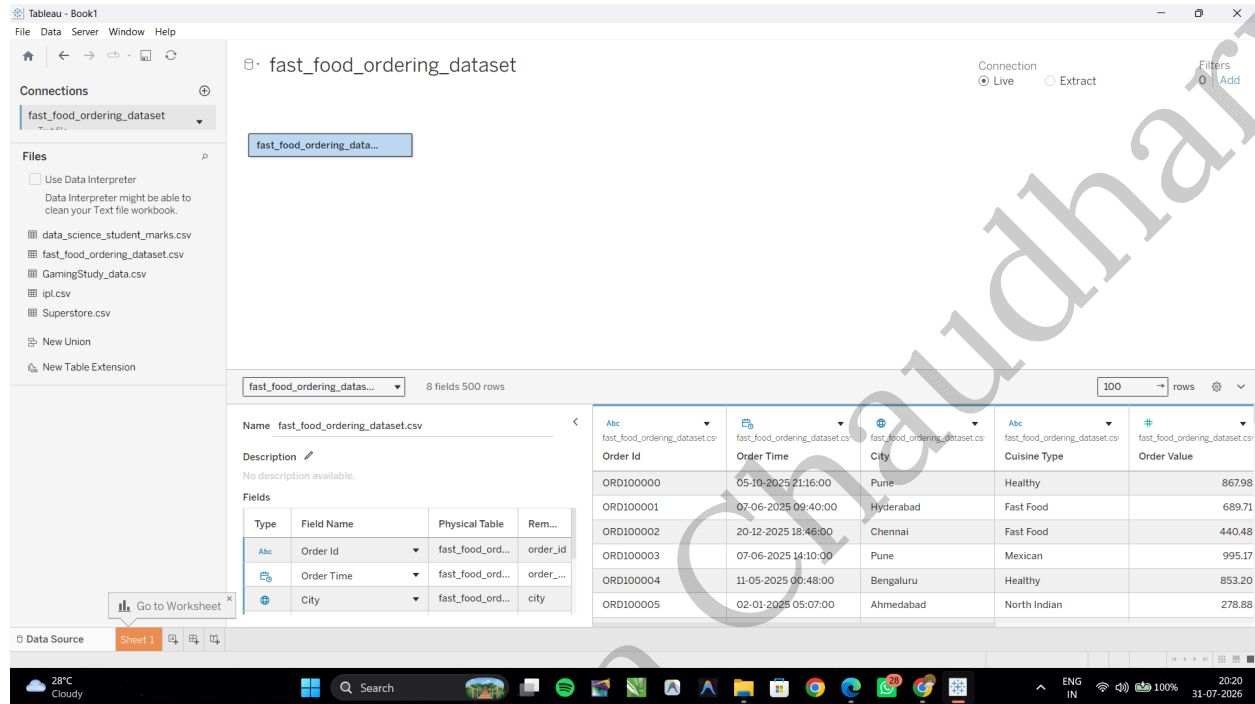
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A) Create Basic Calculated Fields Using Formulas

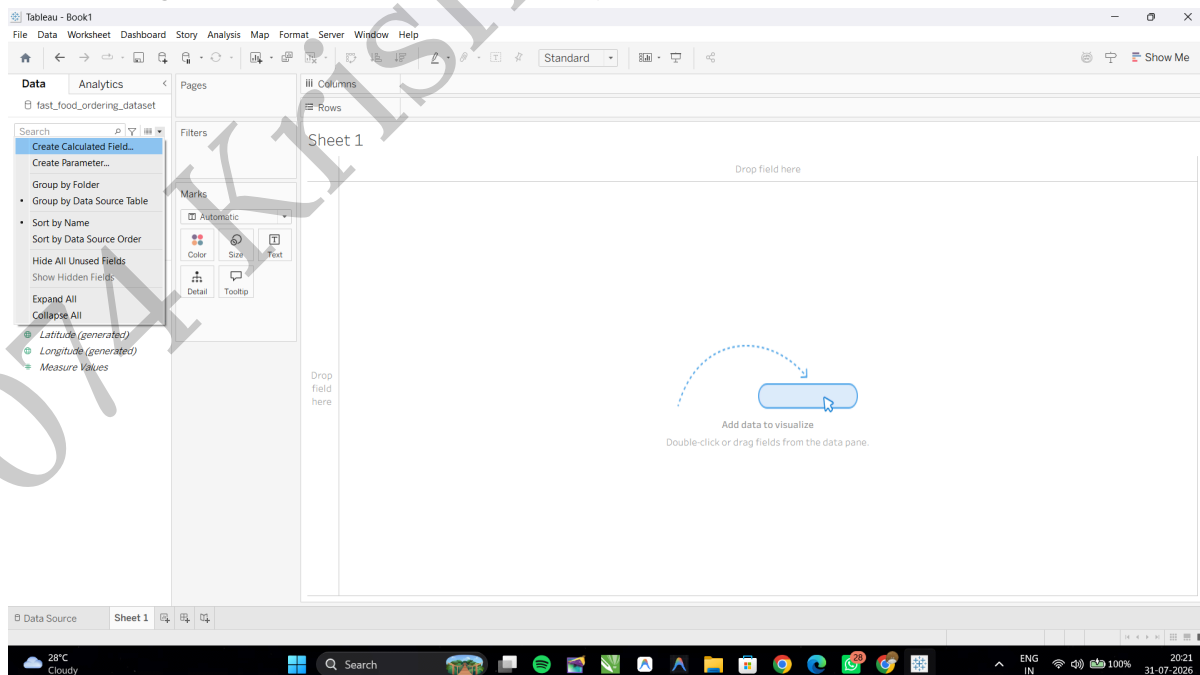
Step 1: Connect Dataset

Import the Dataset: Open Tableau, click **Text File**, select **GamingStudy_data.csv**, click **Open**, then open **Sheet 1**.

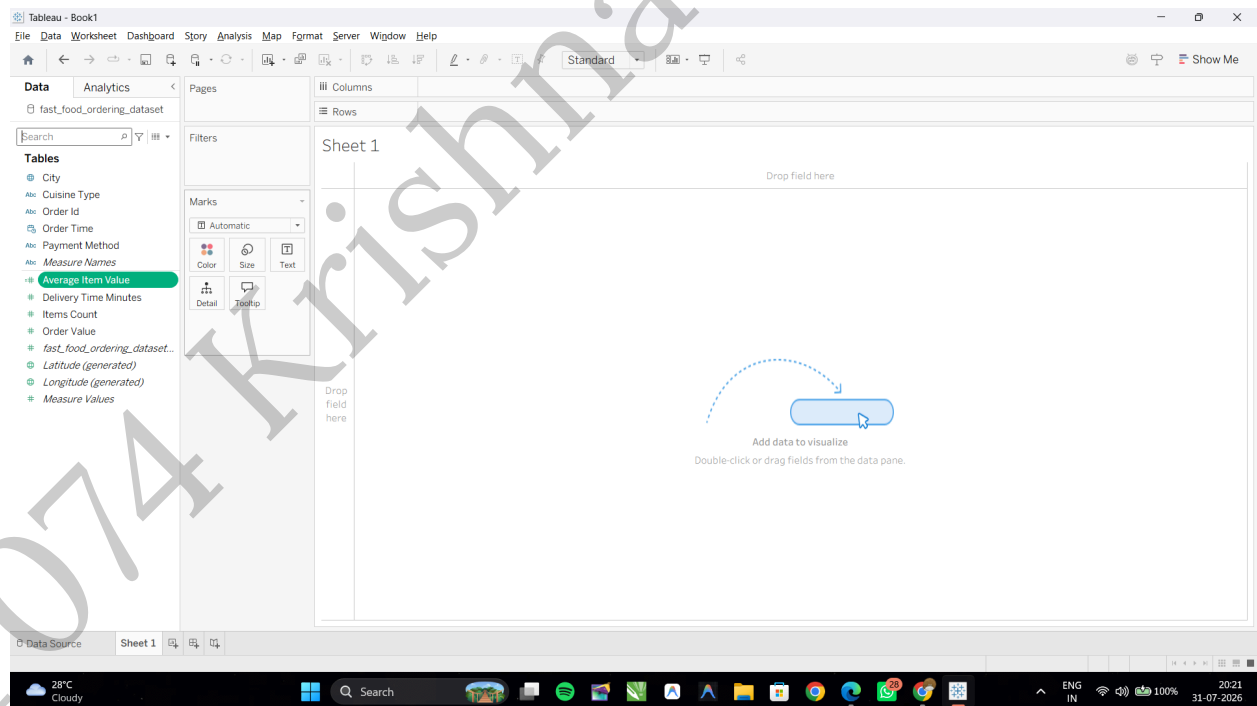
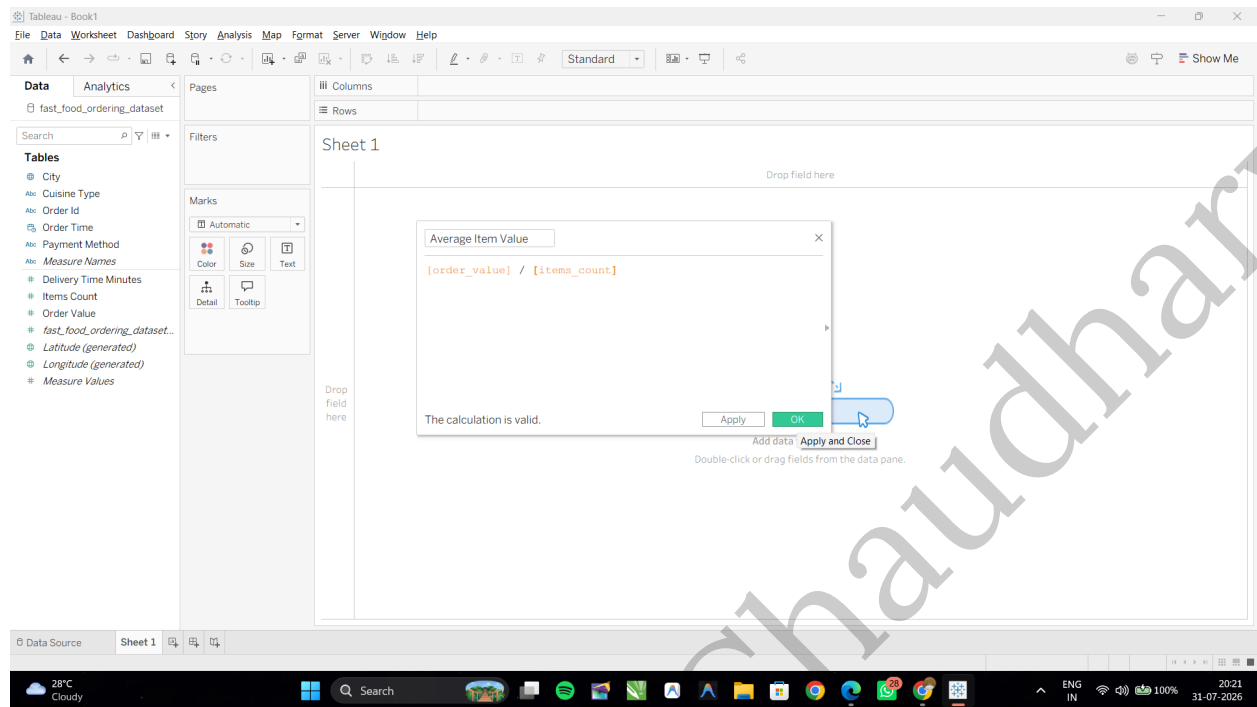


Step 2: Create Basic Calculation

Create a Calculated Field: In the Data Pane, right-click the blank area → **Create Calculated Field**. Name it **Average Item Value**, enter: $[\text{order_value}] / [\text{items_count}]$



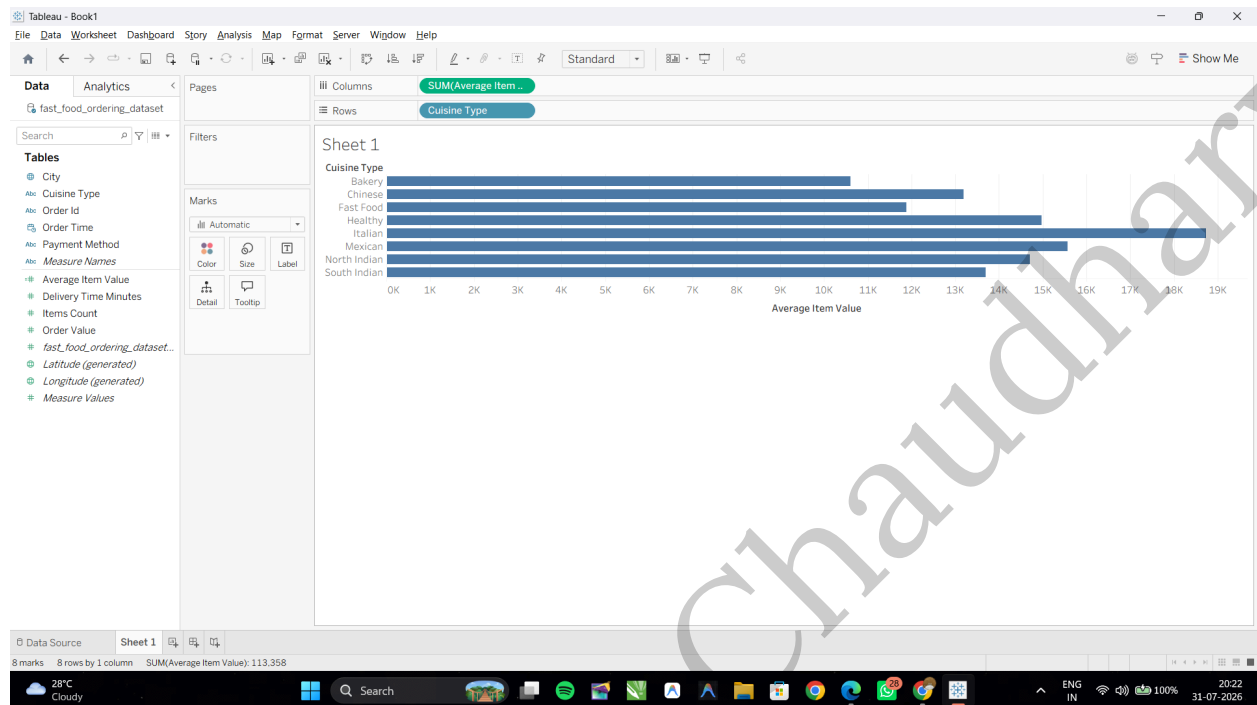
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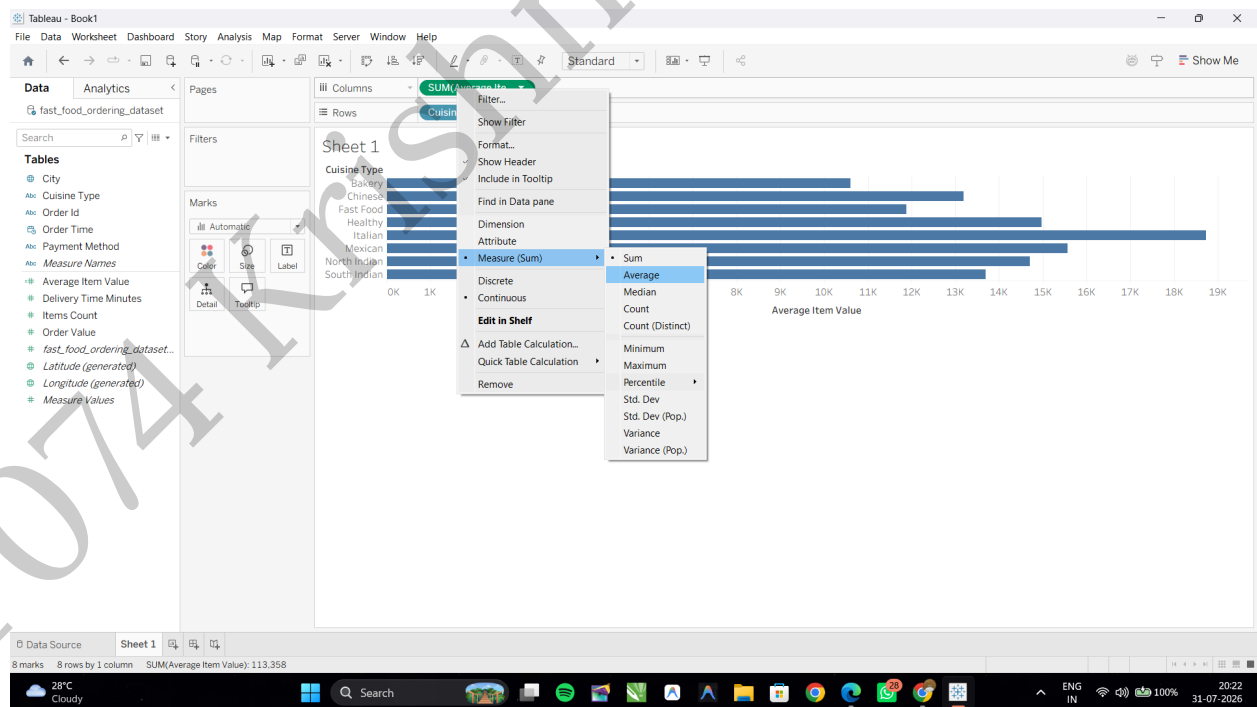
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Create the View: Drag **cuisine_type** to the **Rows** shelf and **Average Item Value** to the **Columns** shelf.



Right-click **Average Item Value** and Measure → Average

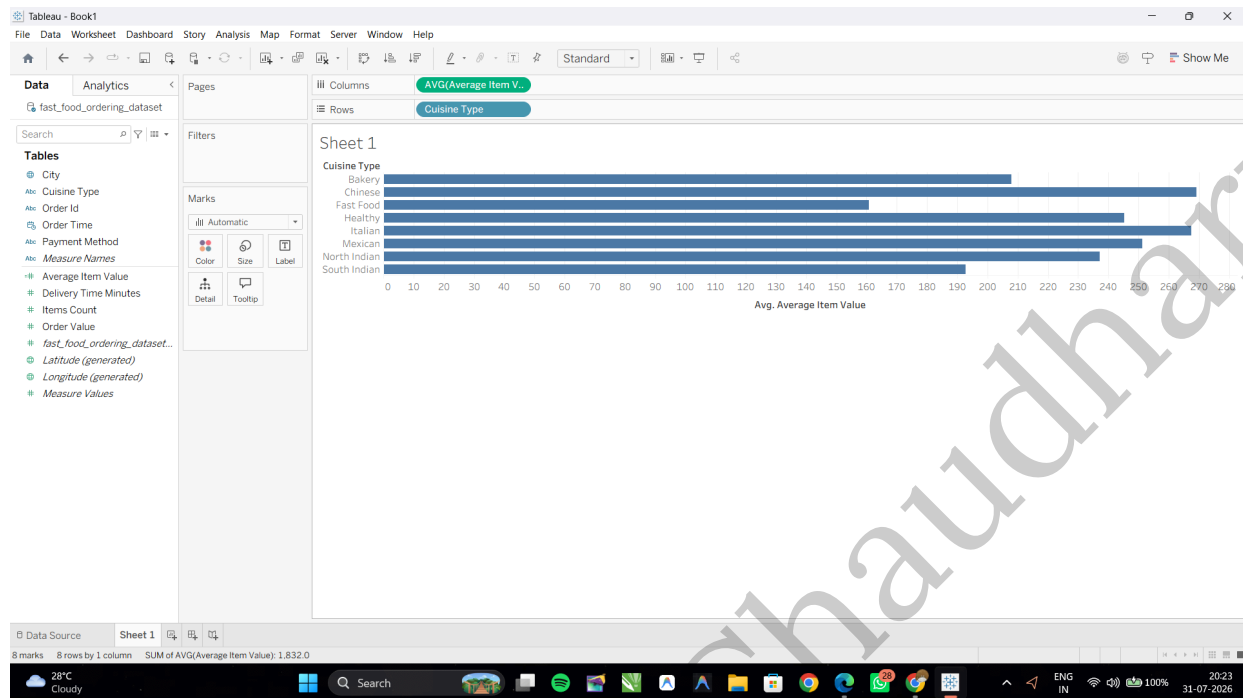


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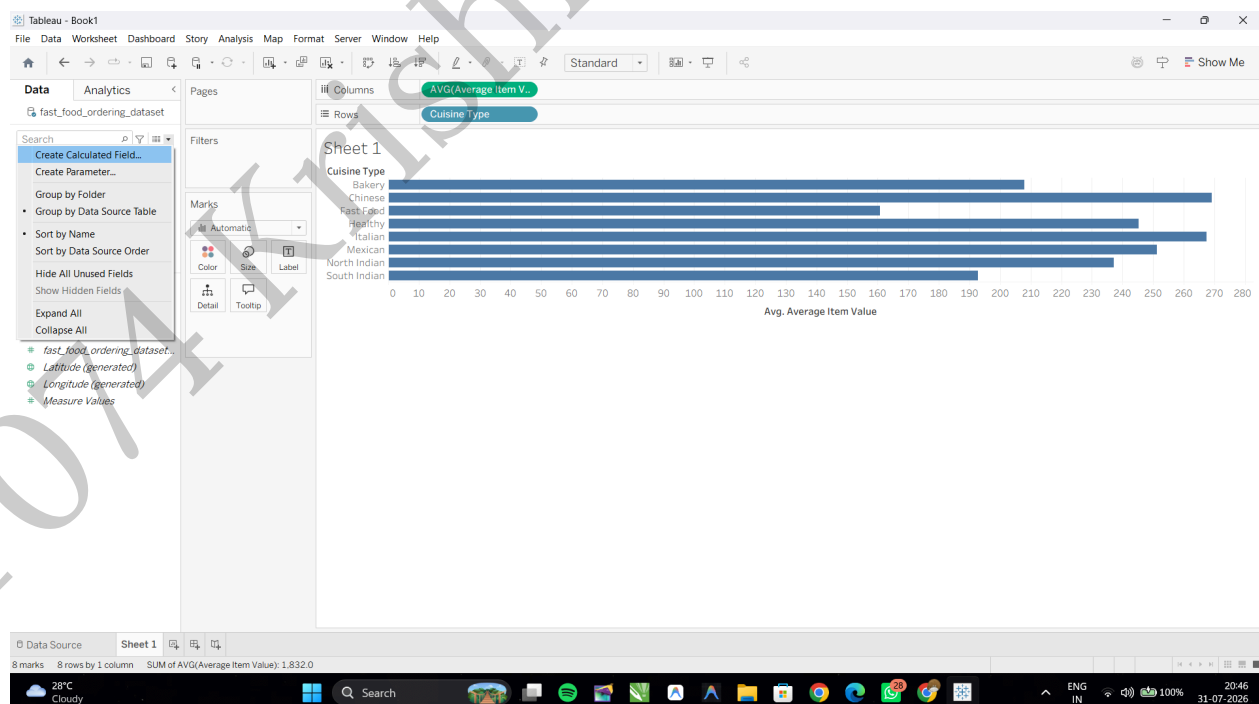
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B) Perform Row-Level Calculations

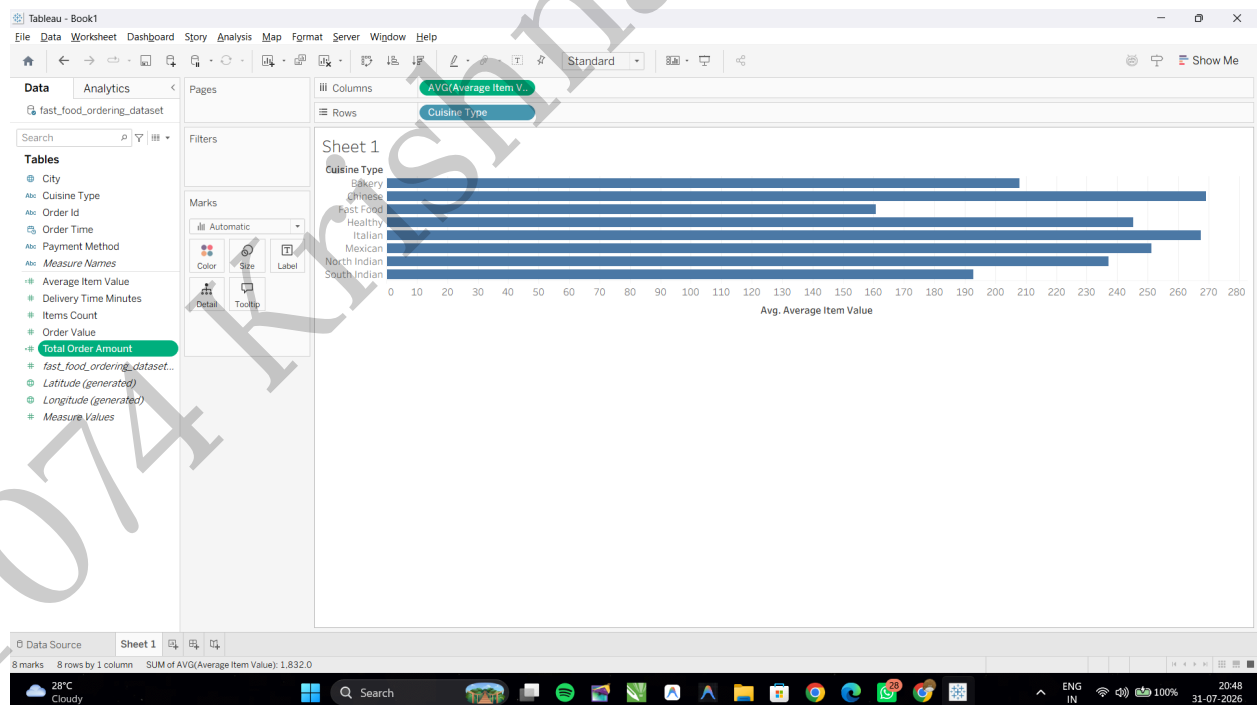
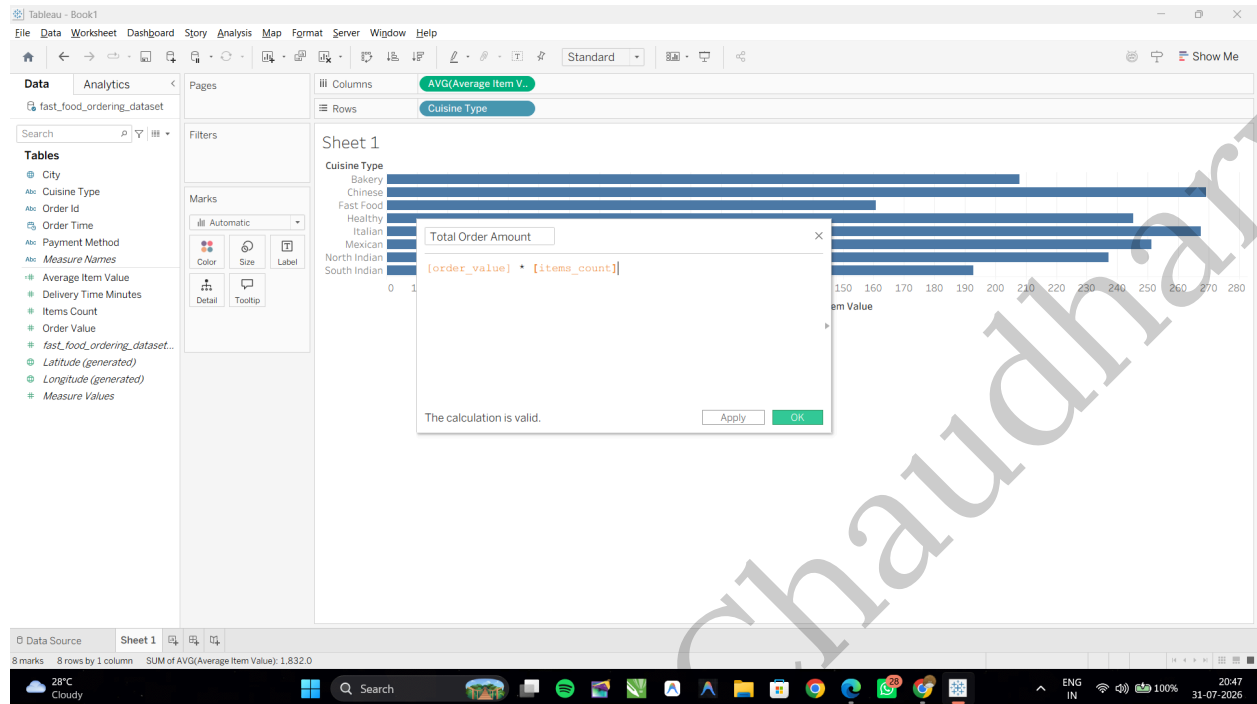
Step 1: Creating a Row-Level Calculated Field

Create a Calculated Field: Right-click the Data Pane → Create Calculated Field, name it Total Order Amount, enter `[order_value] * [items_count]`, verify the calculation is valid, and click OK.



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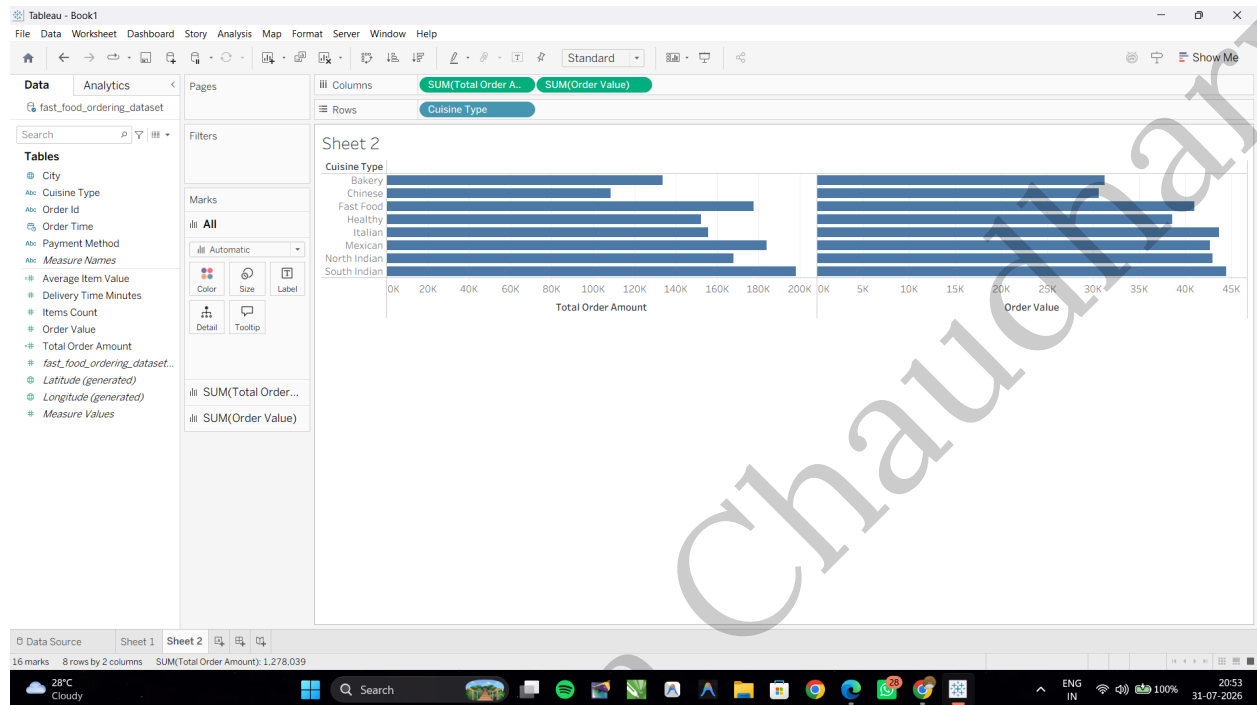


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Step 2: Putting the Field to Use

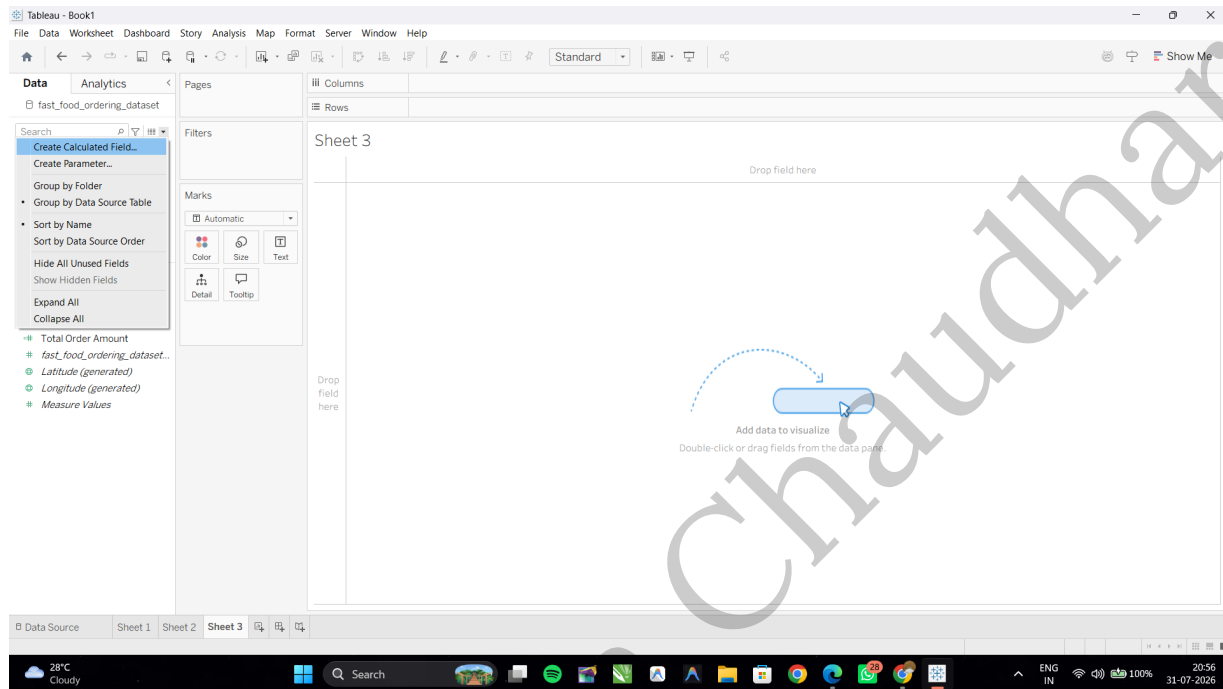
Drag **Cuisine Type** to the **Rows** shelf, then drag **Total Order Amount** and **Order Value** to the **Columns** shelf for comparison.



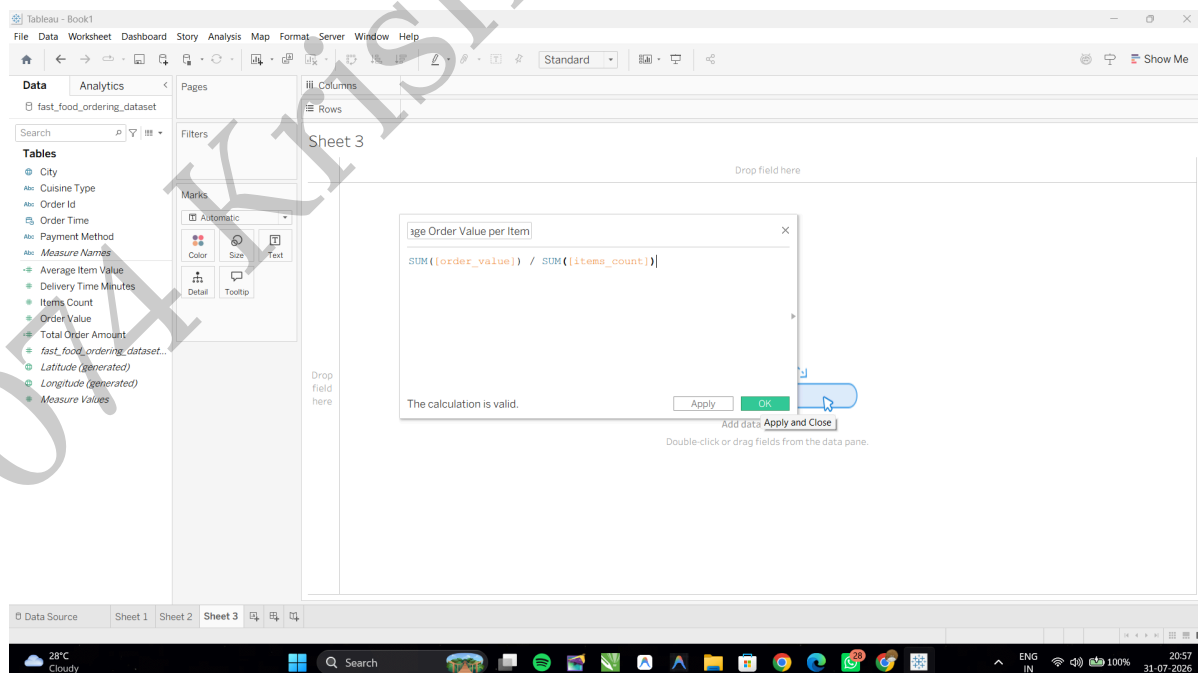
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C: Performing Aggregate-Level Calculations

Step 1: Creating an Aggregate Calculated Field

Right-click the blank area in the Data Pane and select Create Calculated Field.

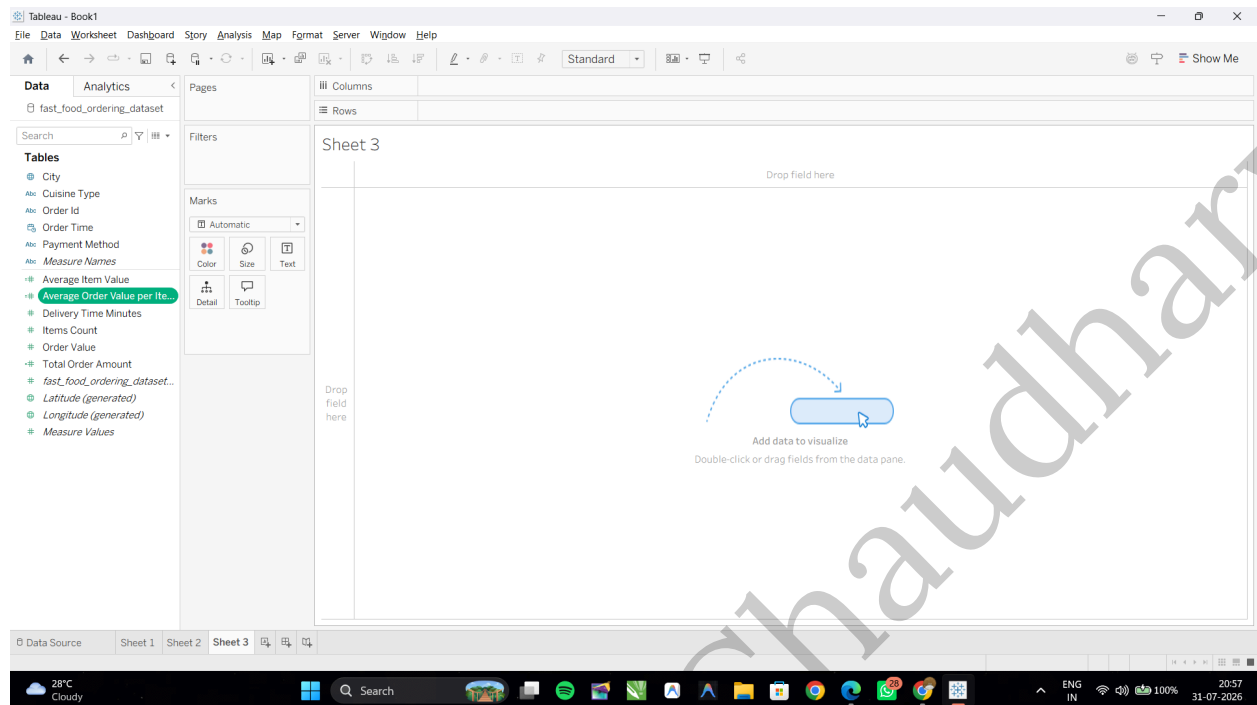


Name the field Average Order Value per Item, enter $\text{SUM}([\text{order_value}]) / \text{SUM}([\text{items_count}])$, then click OK.



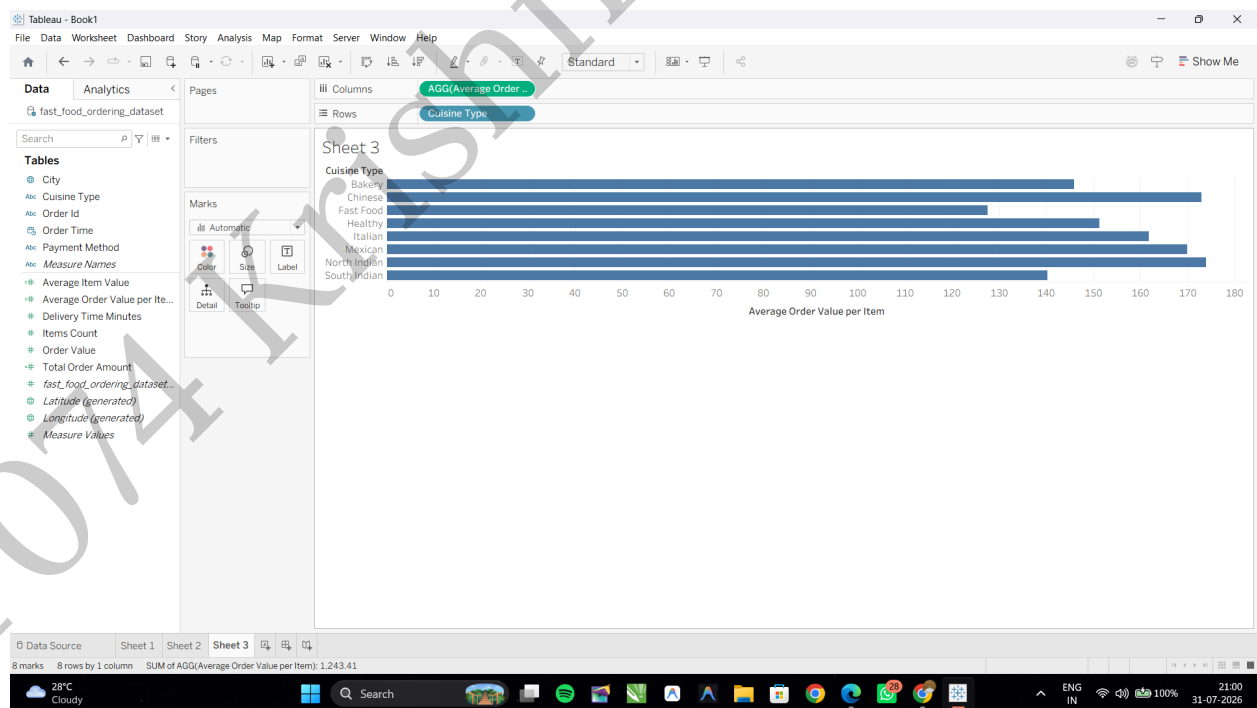
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Step 2: Putting the Field to Use

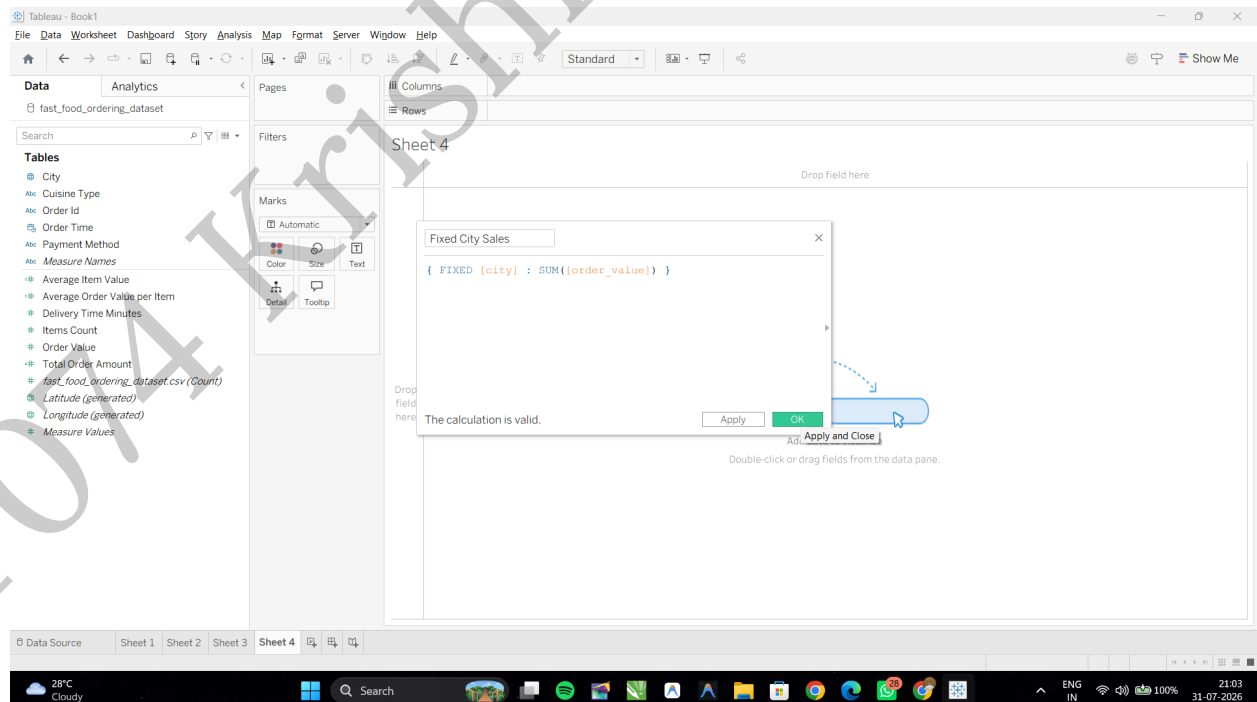
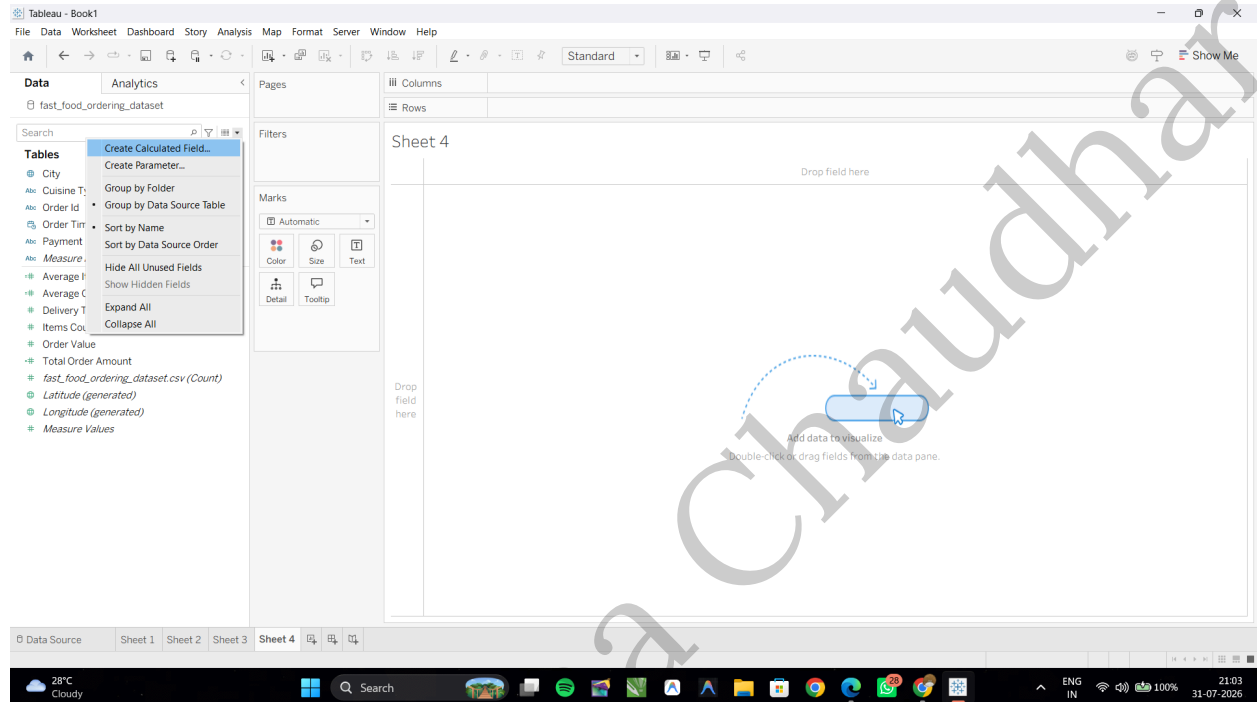
Drag Cuisine Type to the Rows shelf, drag Average Order Value per Item to the Columns shelf, and verify the pill displays as AGG(Average Order Value per Item).



D: Implementing FIXED Level of Detail (LOD) Calculations

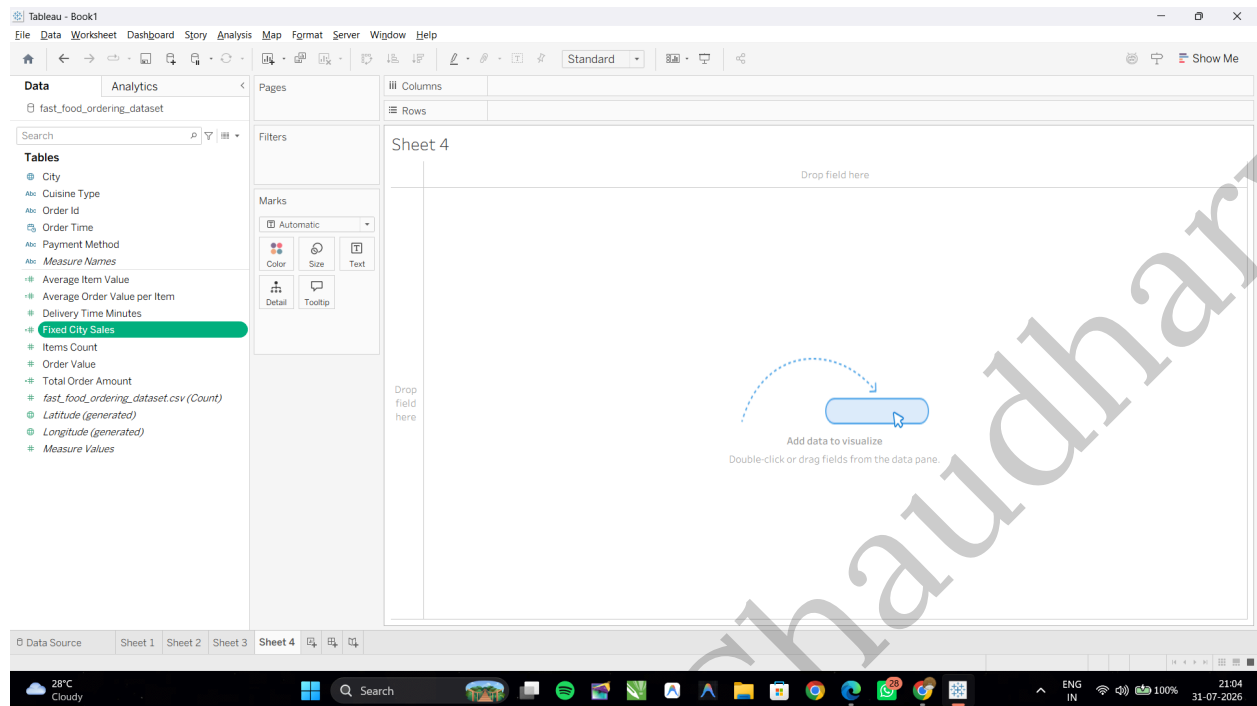
Step 1: Creating a FIXED LOD Calculated Field

Right-click the Data Pane, select Create Calculated Field, name it Fixed City Sales, and enter { FIXED [city] : SUM([order_value]) }.



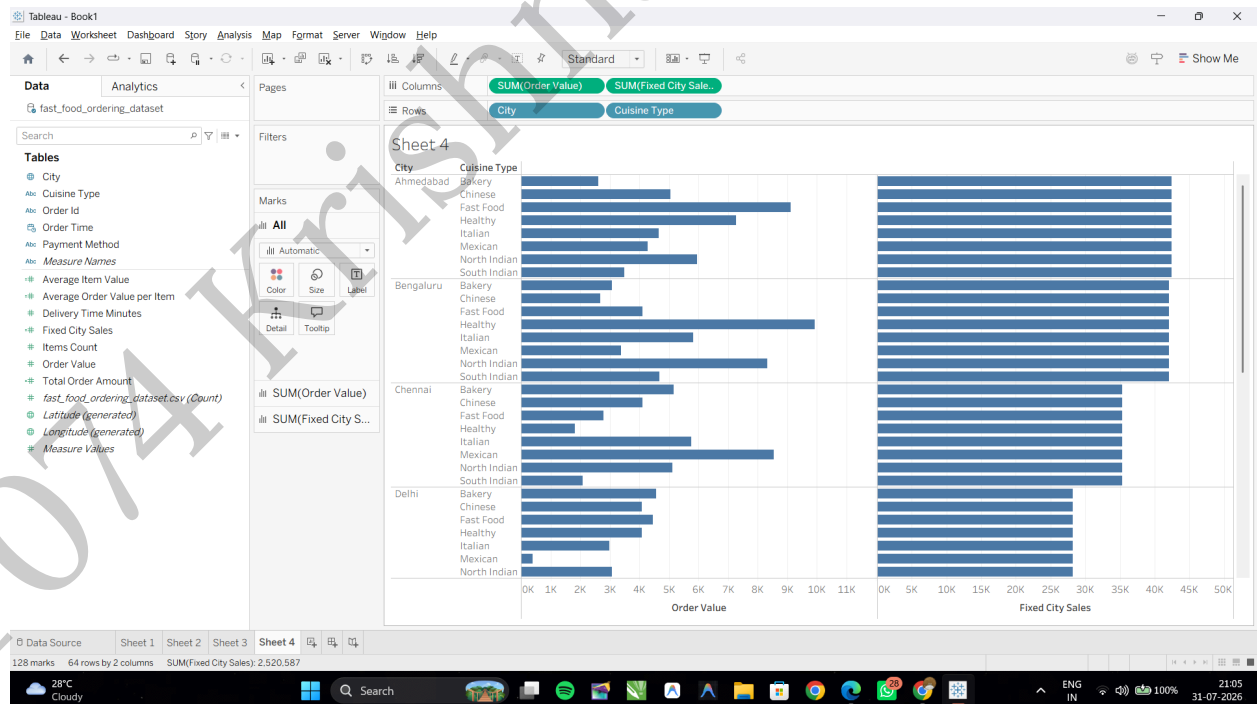
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Step 2: Putting the Field to Use

Drag **City** and **Cuisine Type** to the **Rows** shelf, then drag **Order Value** and **Fixed City Sales** to the **Columns** shelf.



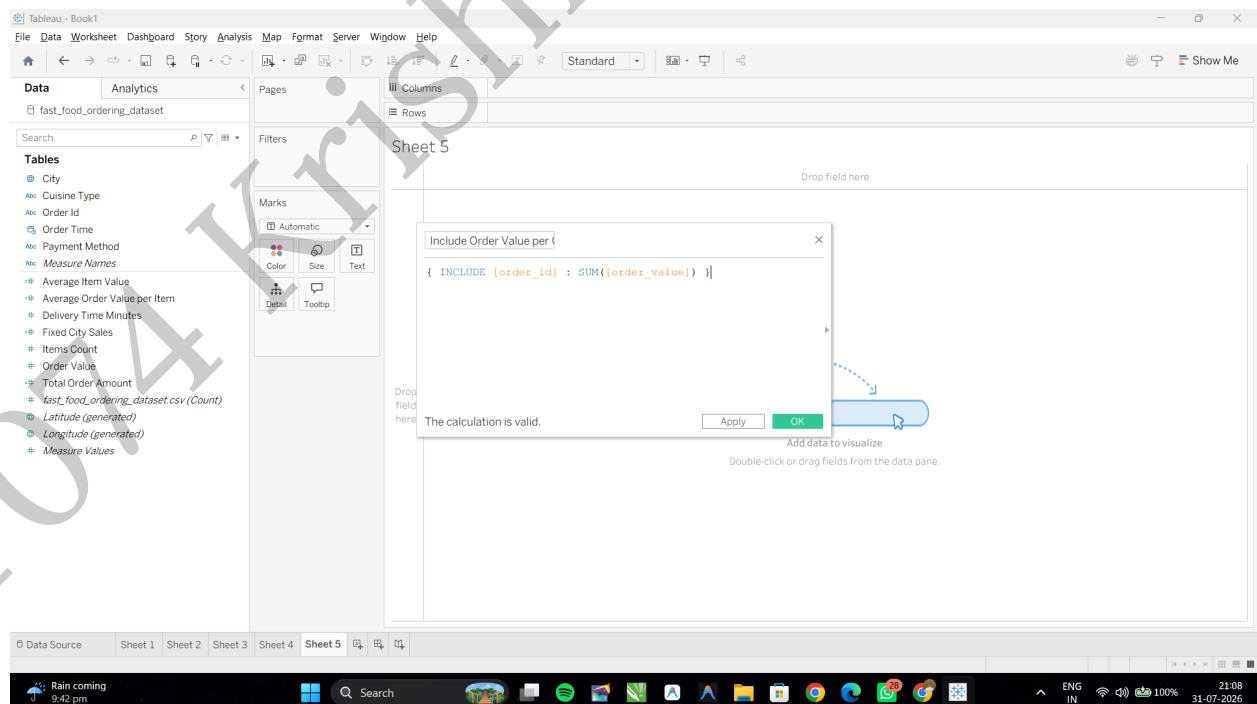
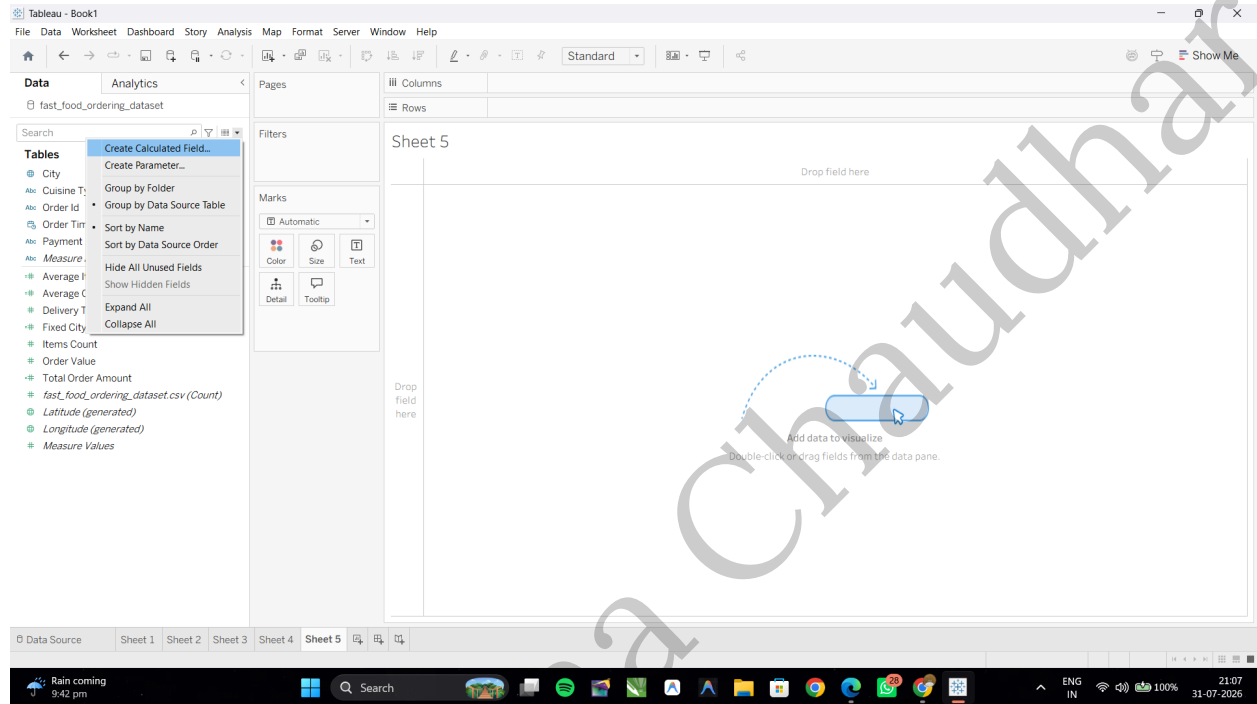
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E: Using INCLUDE and EXCLUDE LOD Expressions

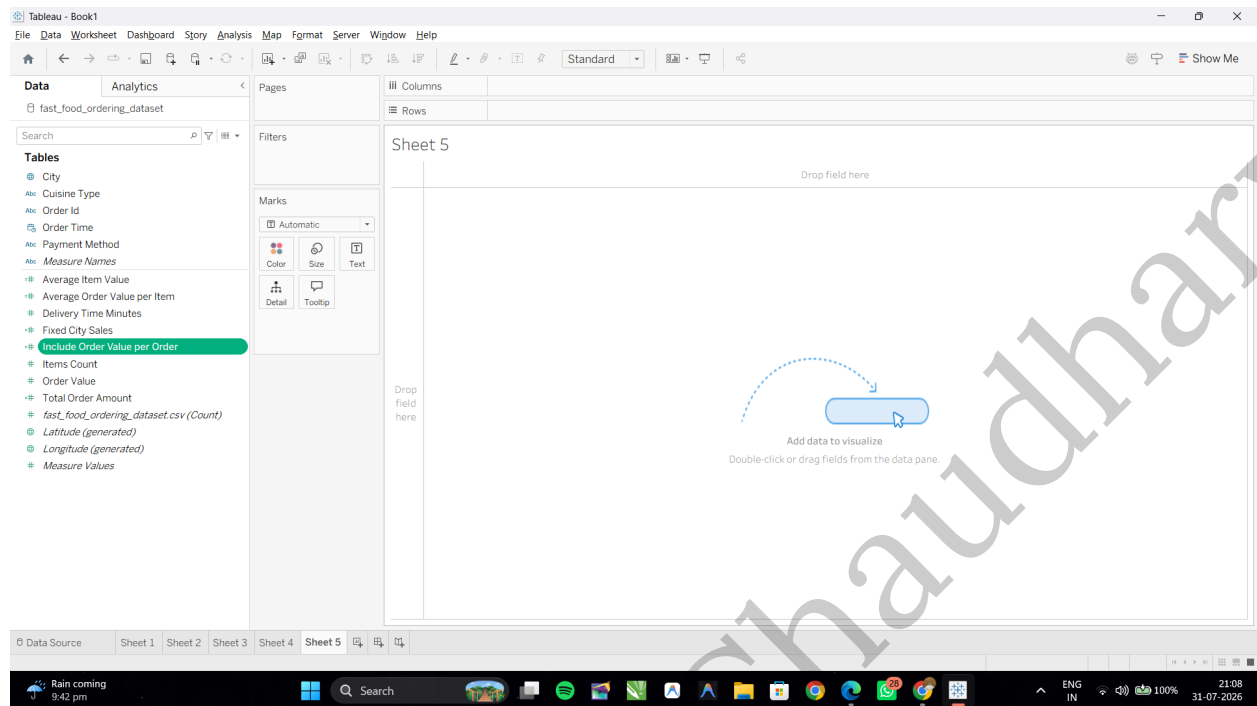
Step 1: Creating an INCLUDE LOD Calculated Field

Right-click the Data Pane, select Create Calculated Field, name it Include Order Value per Order, and enter { INCLUDE [order_id] : SUM([order_value]) }.



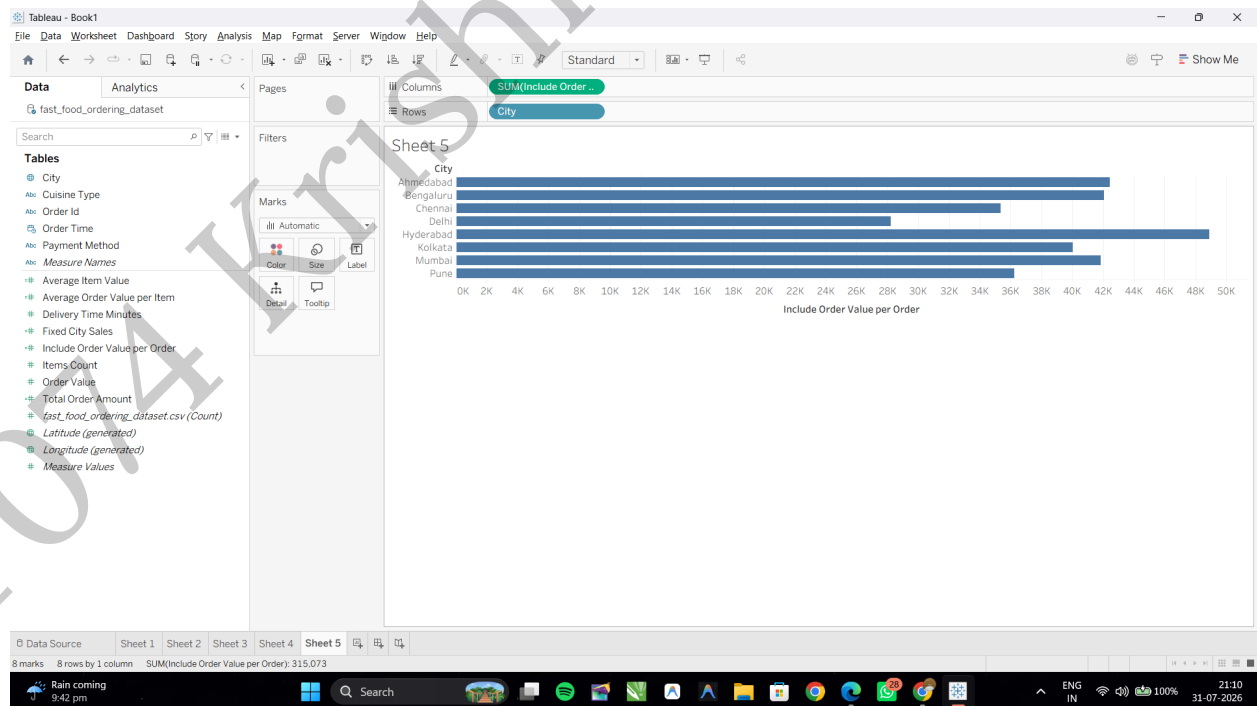
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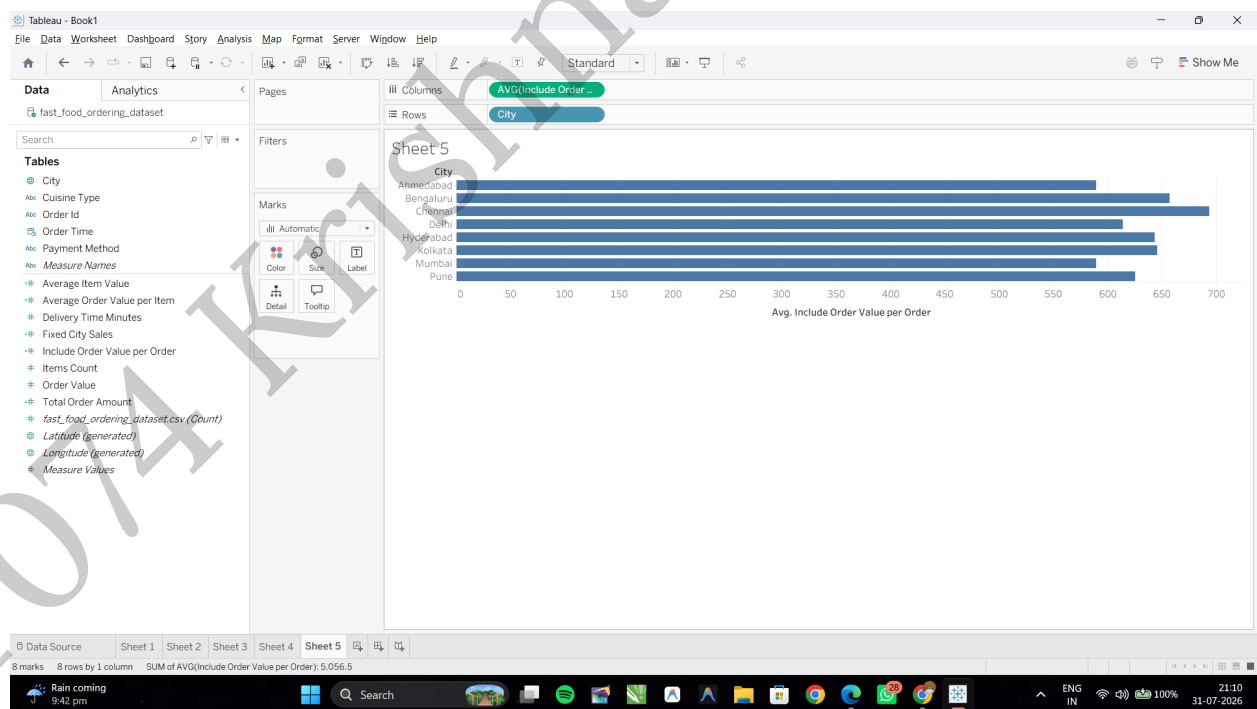
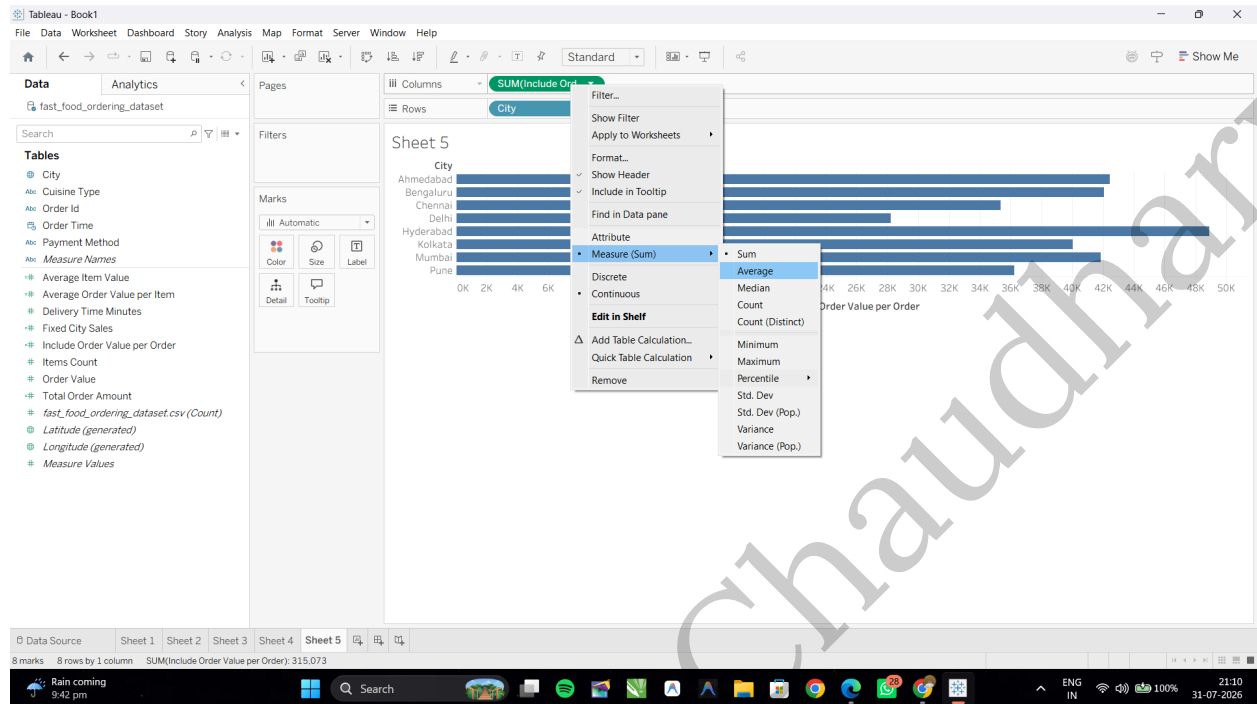
Step 2: Putting the INCLUDE Field to Use

Drag City to the Rows shelf, drag Include Order Value per Order to the Columns shelf, then change its aggregation from Sum to Average.



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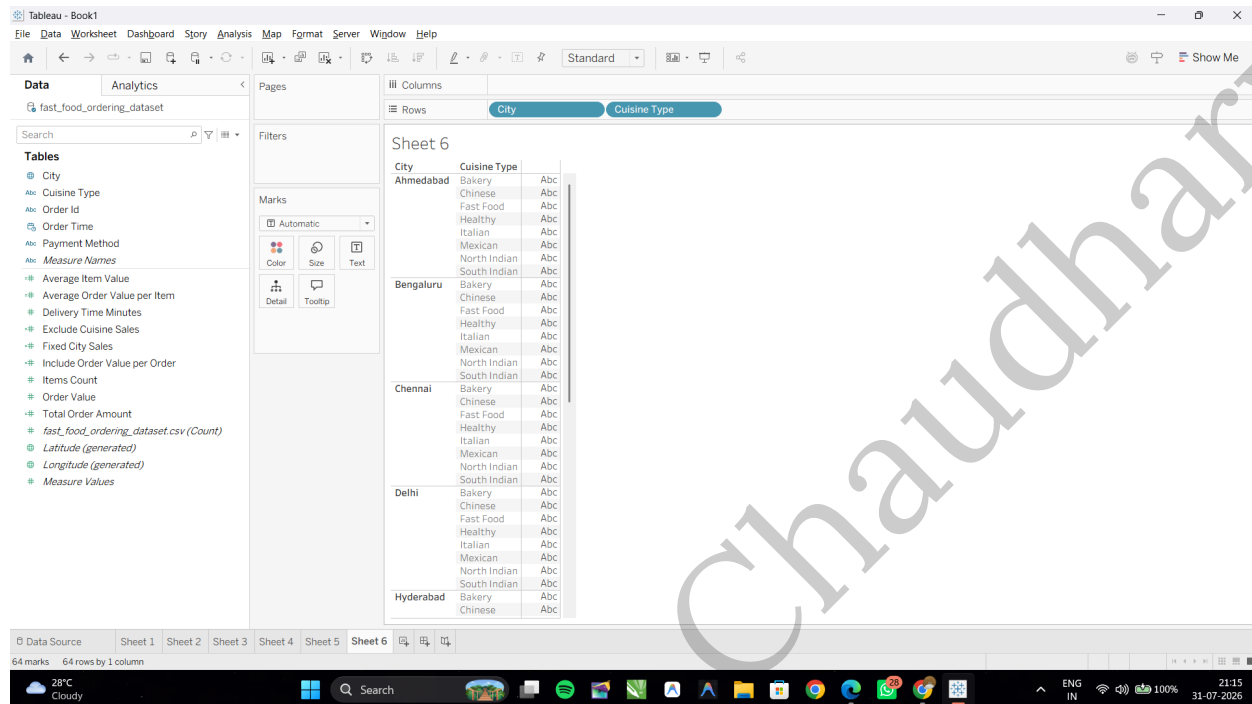


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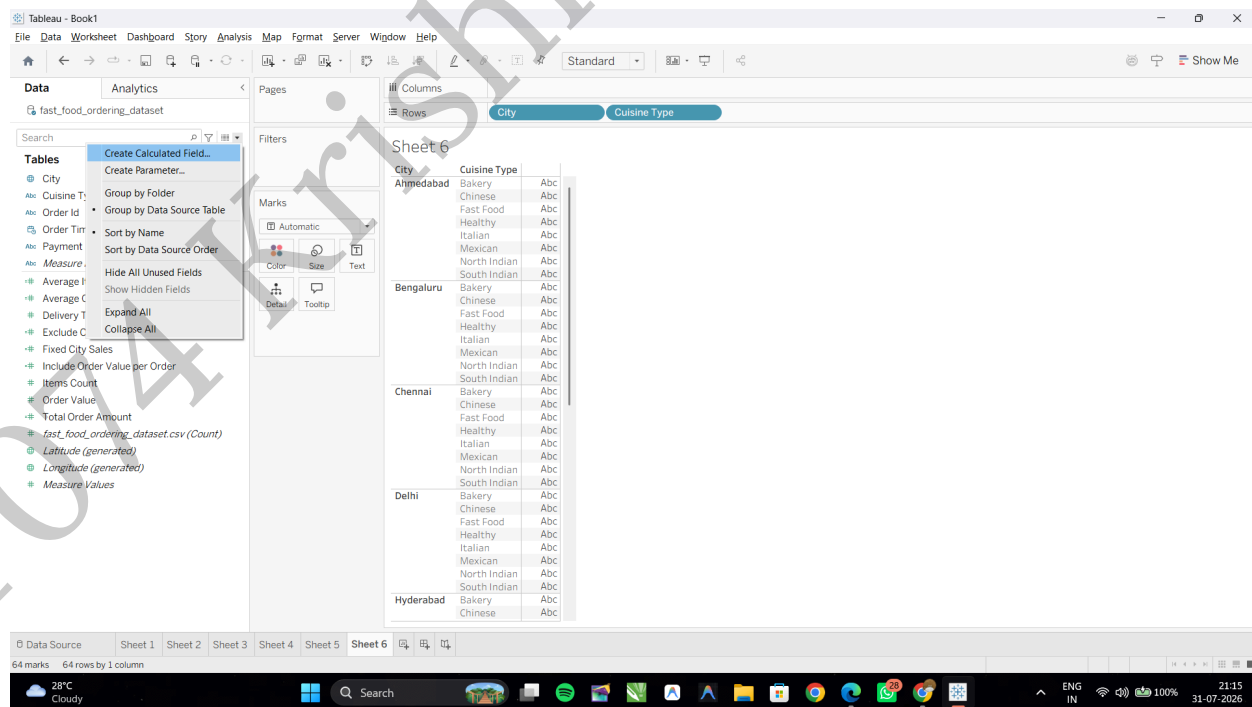
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Step 3: Creating an EXCLUDE LOD Calculated Field

Open a new worksheet, then drag City and Cuisine Type to the Rows shelf.

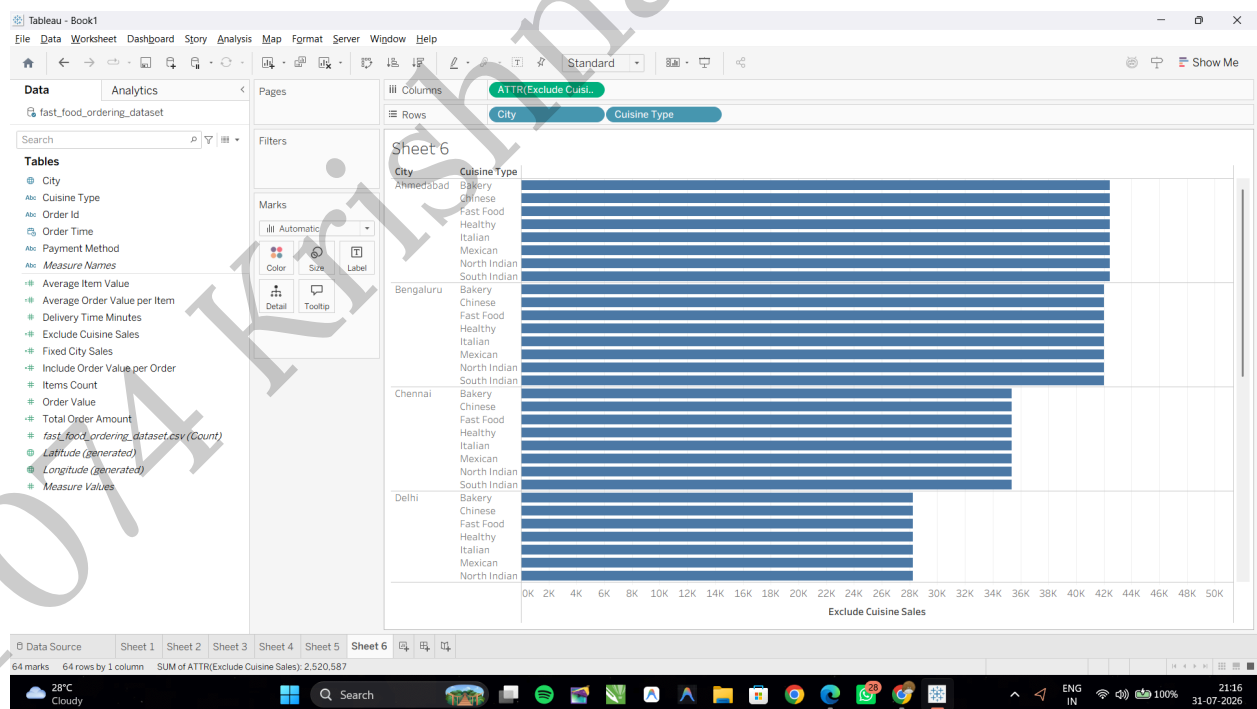
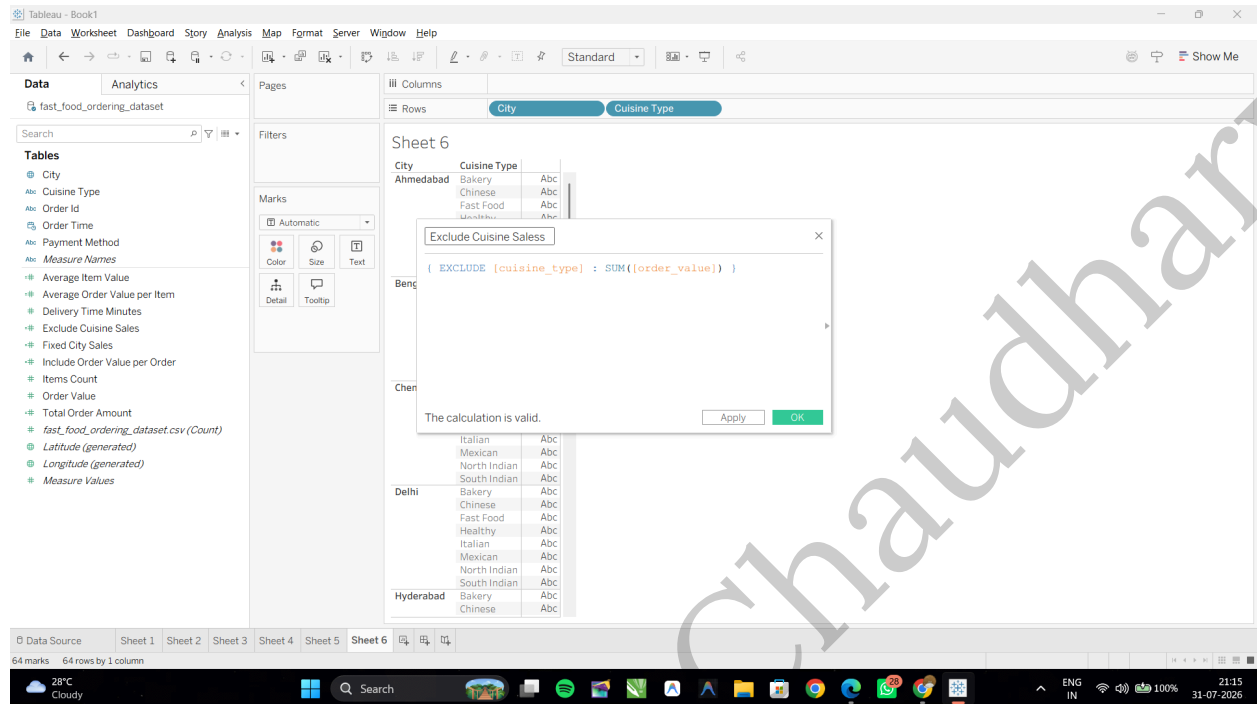


Right-click the Data Pane, select Create Calculated Field, name it Exclude Cuisine Sales, enter { **EXCLUDE [cuisine_type] : SUM([order_value])** }, click OK, then drag it to the Columns shelf.



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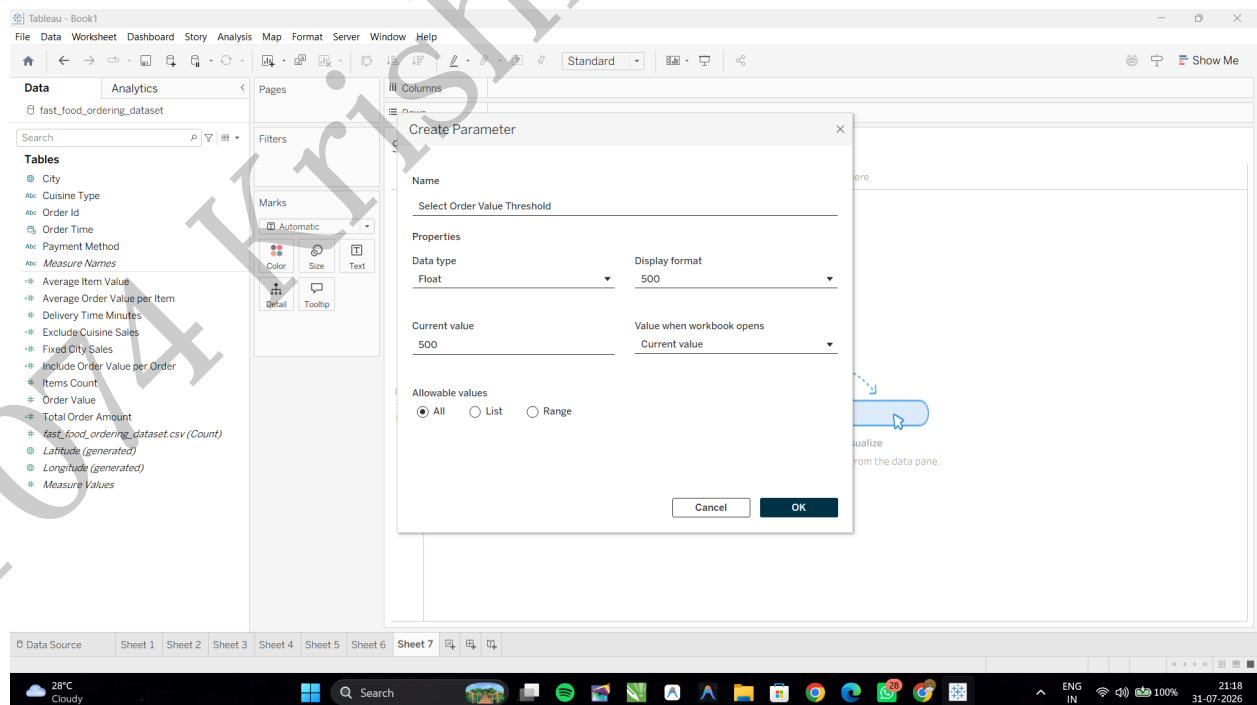
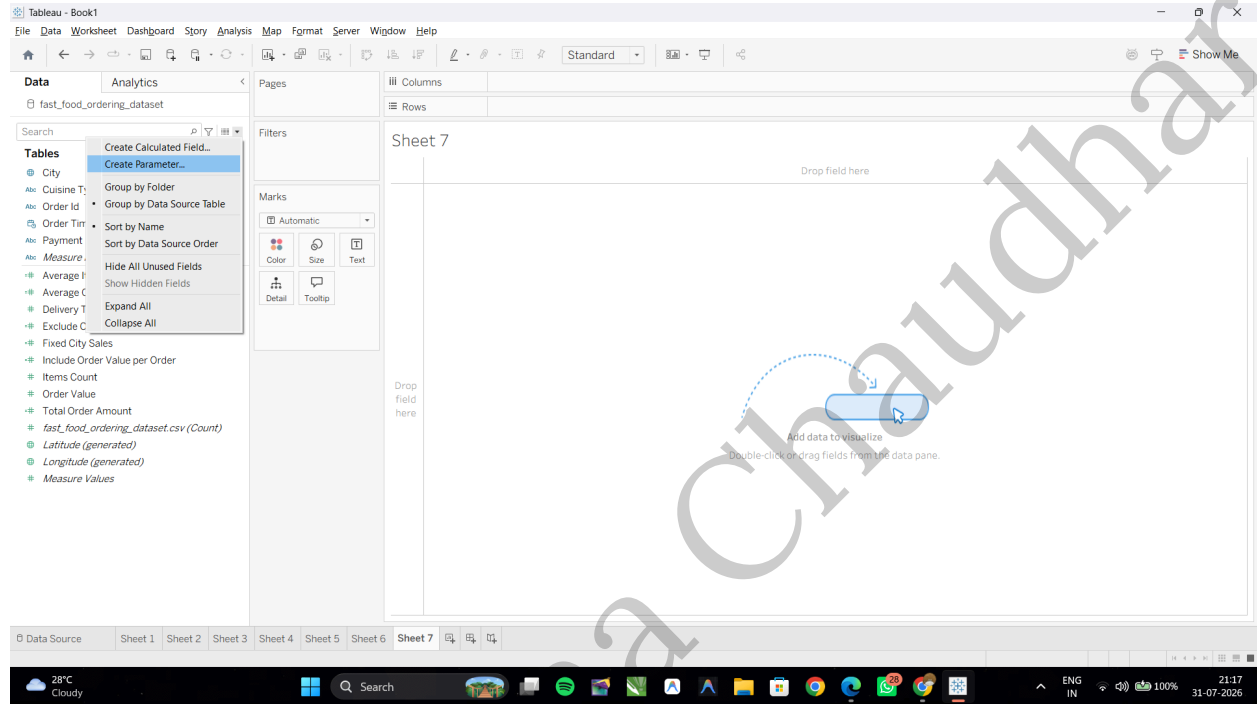
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F: Creating and Using Parameters for Dynamic Analysis

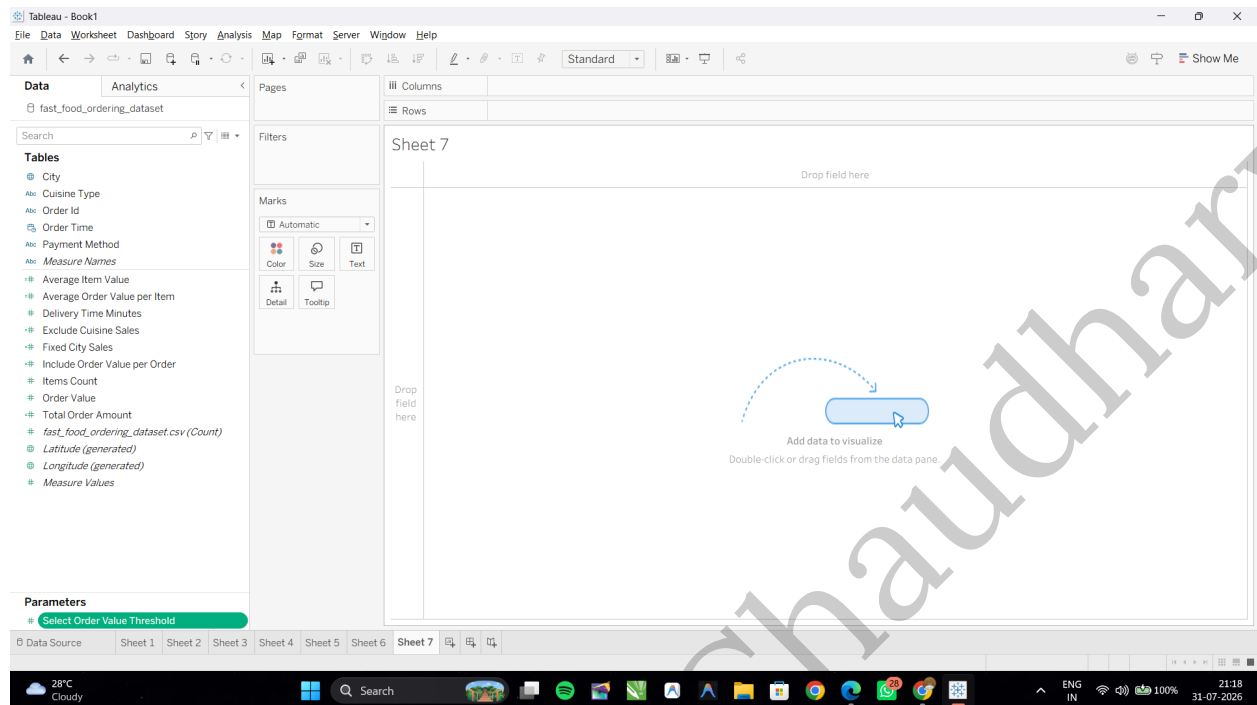
Step 1: Creating a Parameter

Go to the Data Pane, select Create Parameter, name it Select Order Value Threshold, set Data Type to Float, Current Value to 500, keep Allowable Values as All, then click OK.

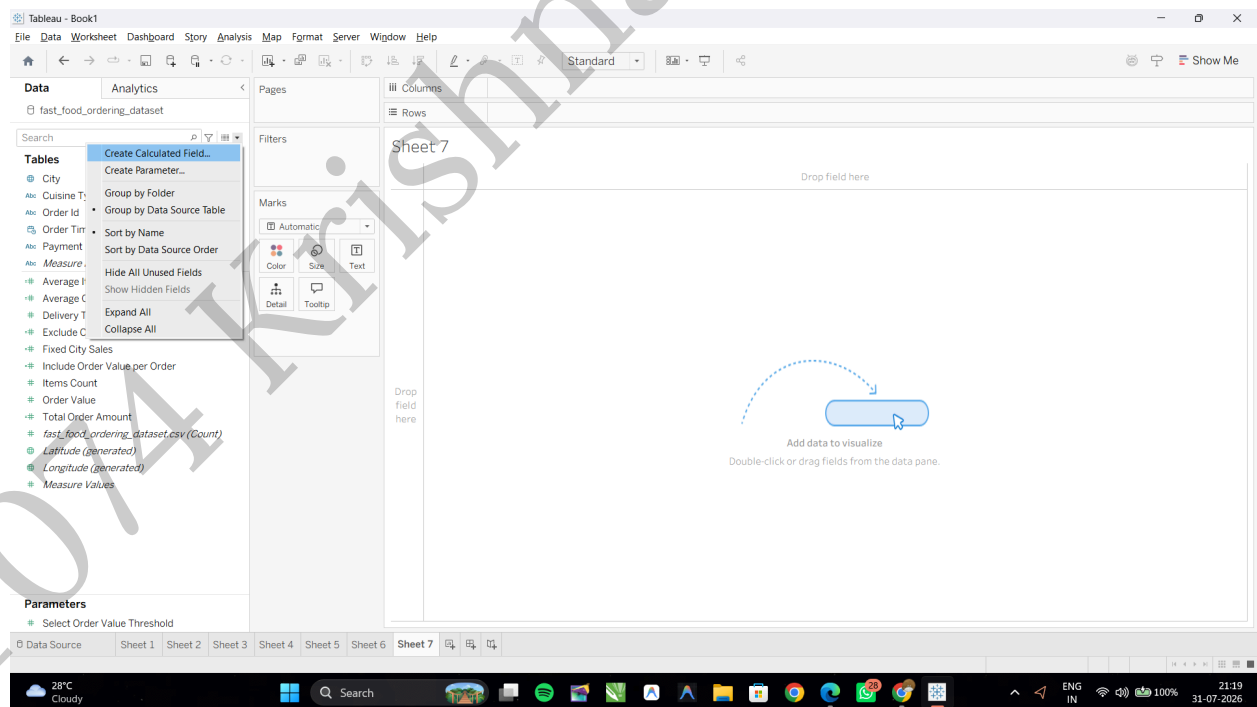


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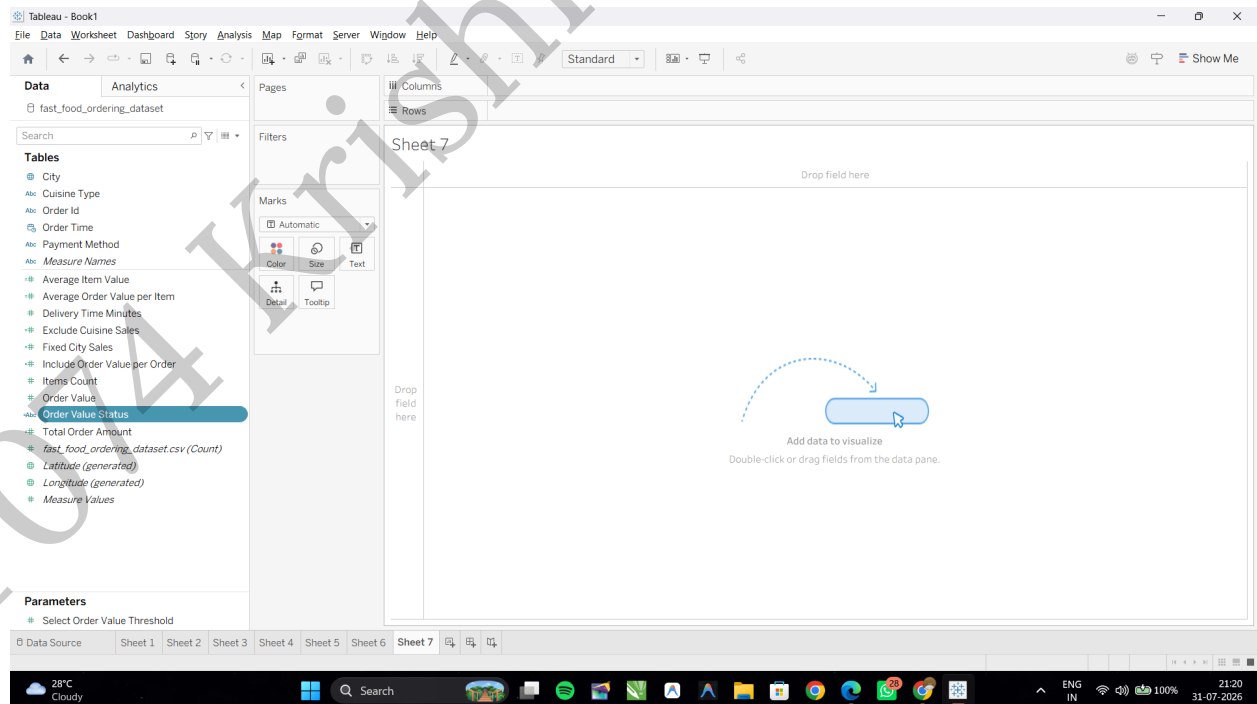
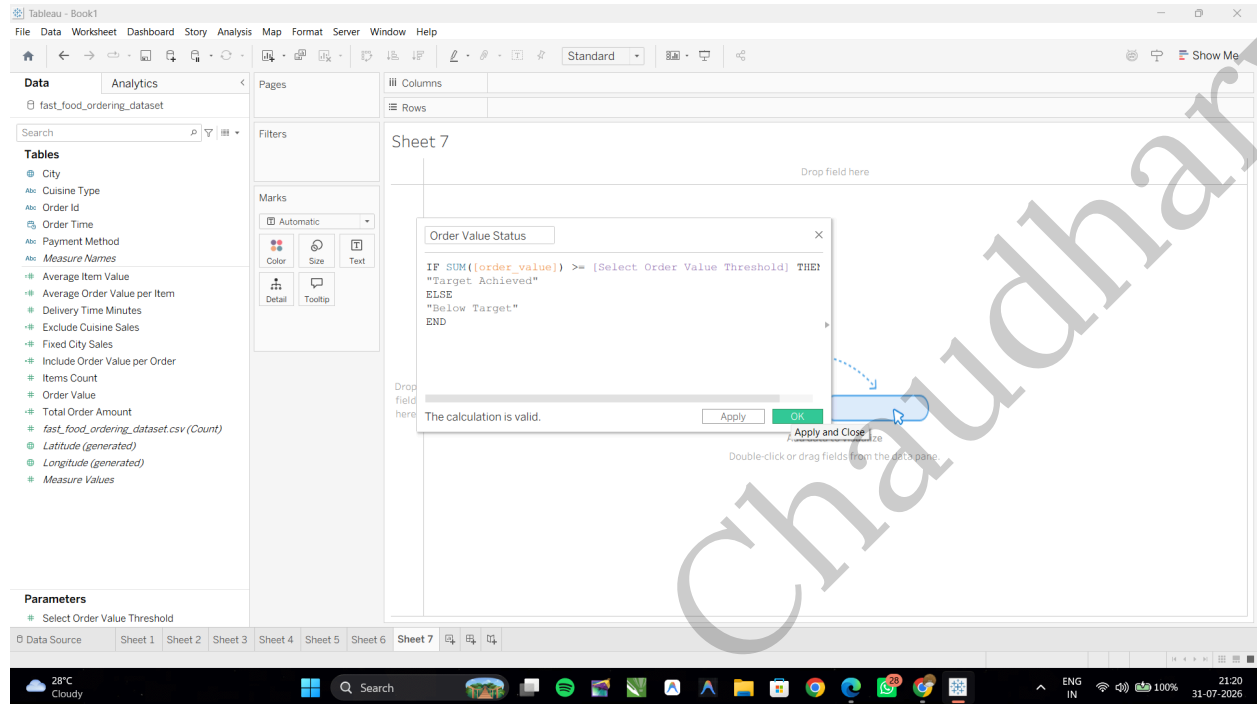
Step 2: Linking the Parameter to a Calculated Field



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Right-click the Data Pane, select Create Calculated Field, name it Order Value Status, enter **IF SUM([order_value]) >= [Select Order Value Threshold] THEN "Target Achieved" ELSE "Below Target" END**, verify the calculation is valid, and click OK.

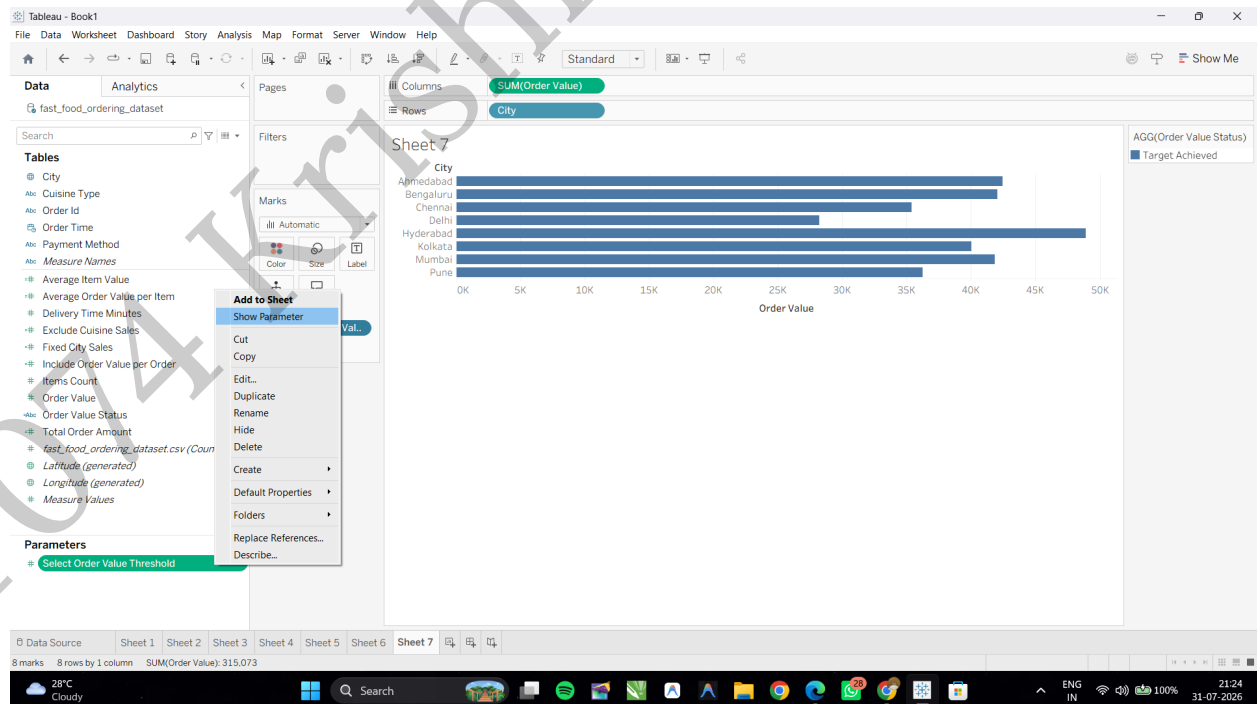
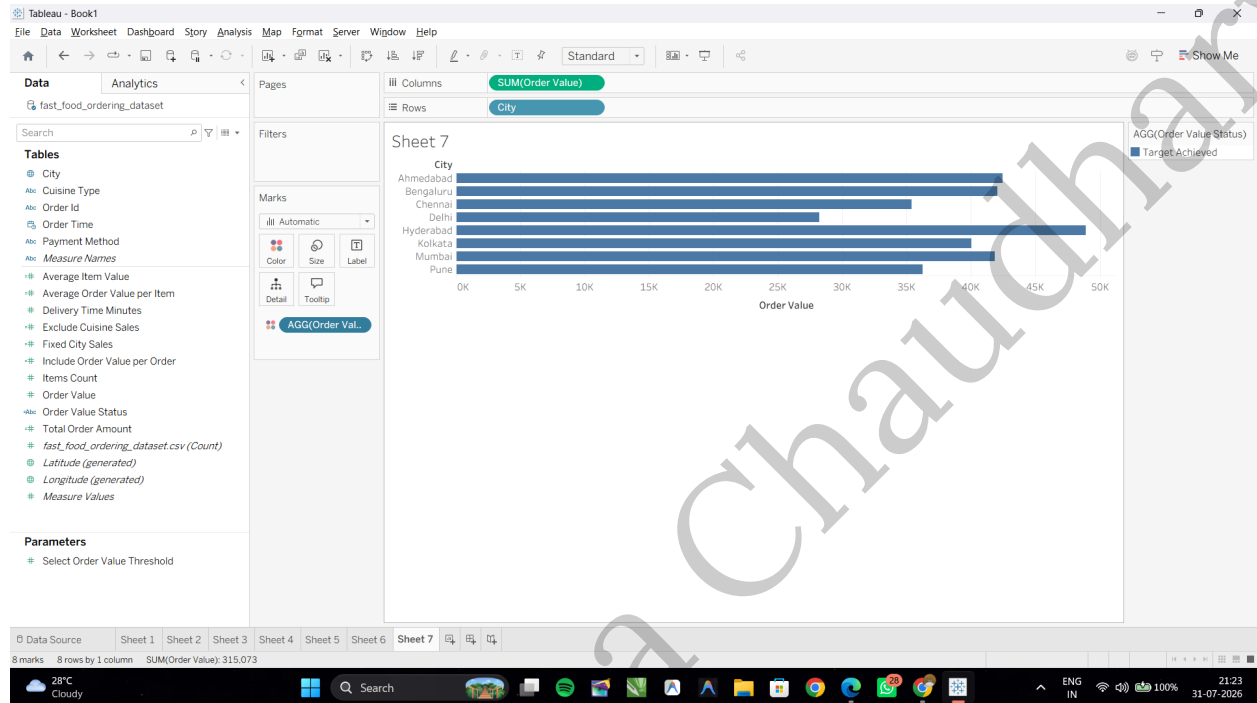


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Step 3: Putting the Interactive Parameter to Use

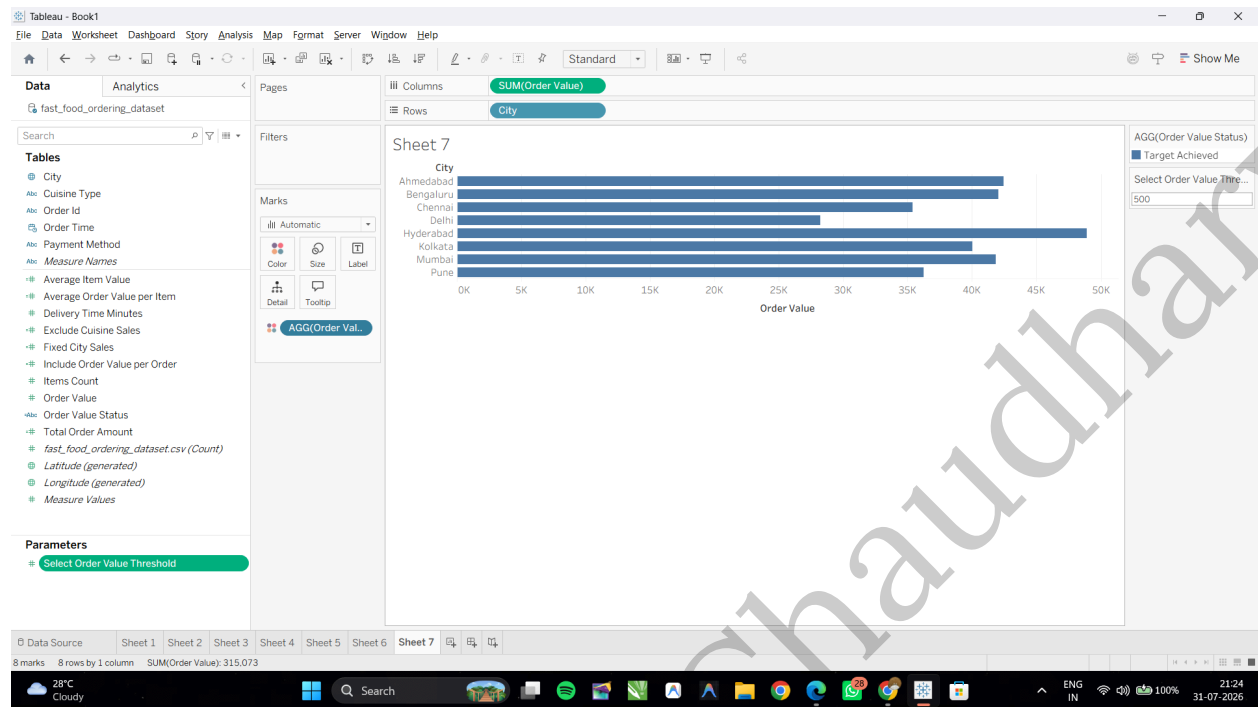
Drag City to the Rows shelf, Order Value to the Columns shelf, Order Value Status to Color, then Show Parameter for Select Order Value Threshold and adjust its value to see the bar colors update automatically.



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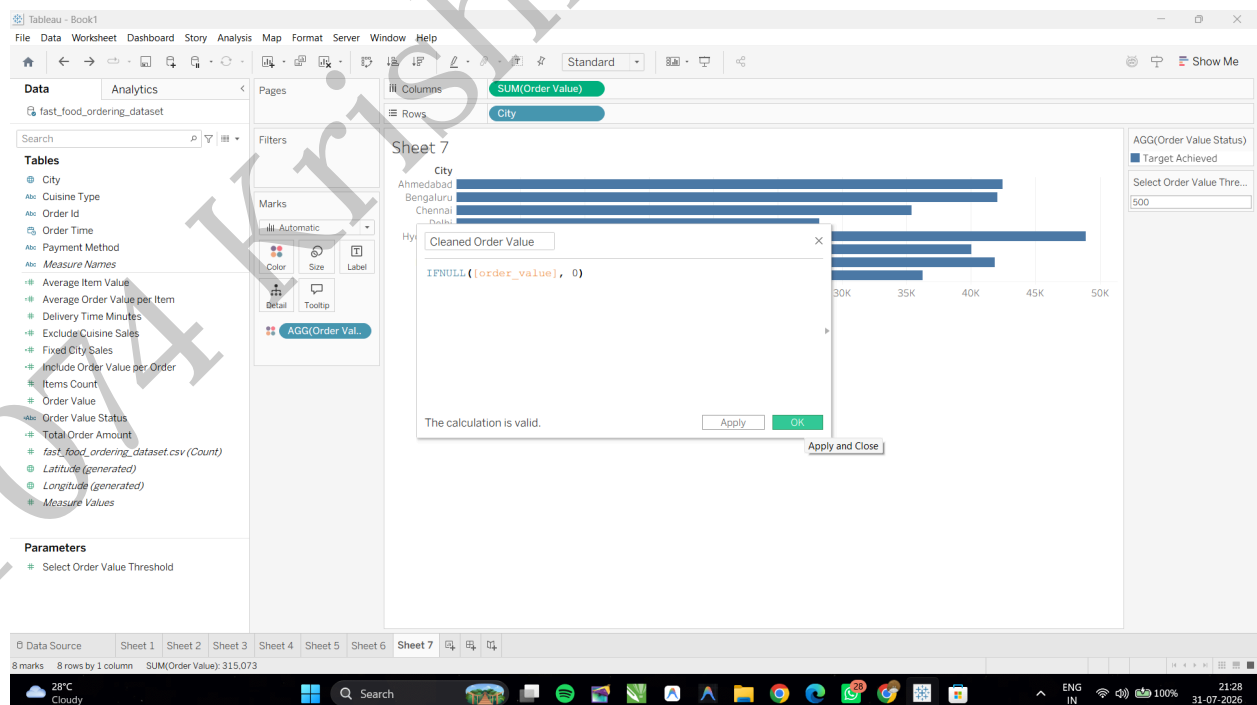
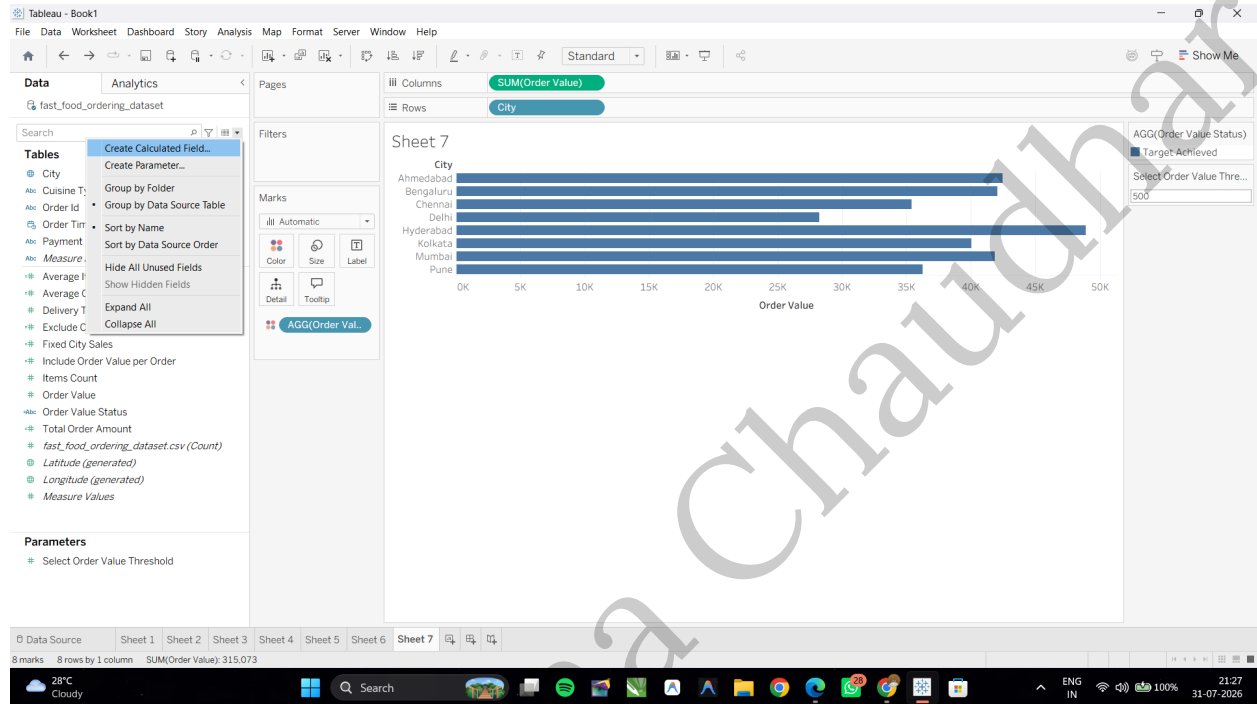
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G: Apply Calculations to Fix Data Issues and Enhance Insights

Step 1: Handling Null or Missing Values with IFNULL

Right-click the Data Pane, select Create Calculated Field, name it Cleaned Order Value, enter `IFNULL([order_value], 0)`, verify The calculation is valid, and click OK.

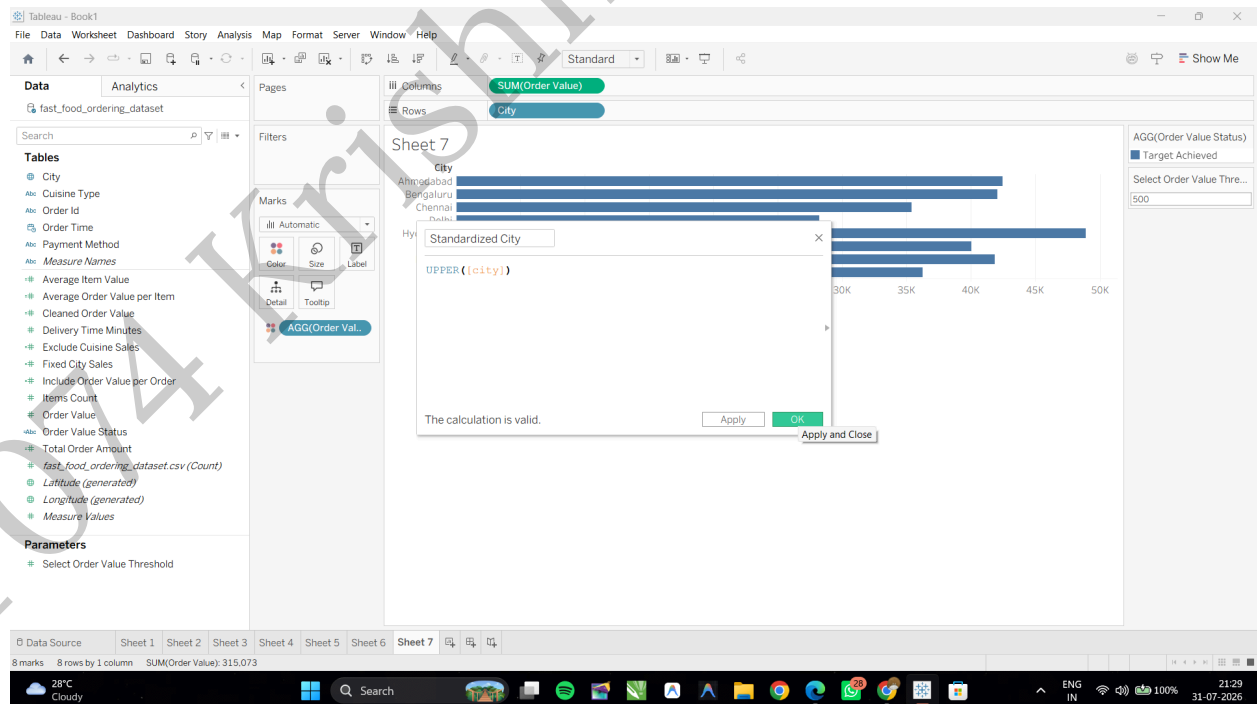
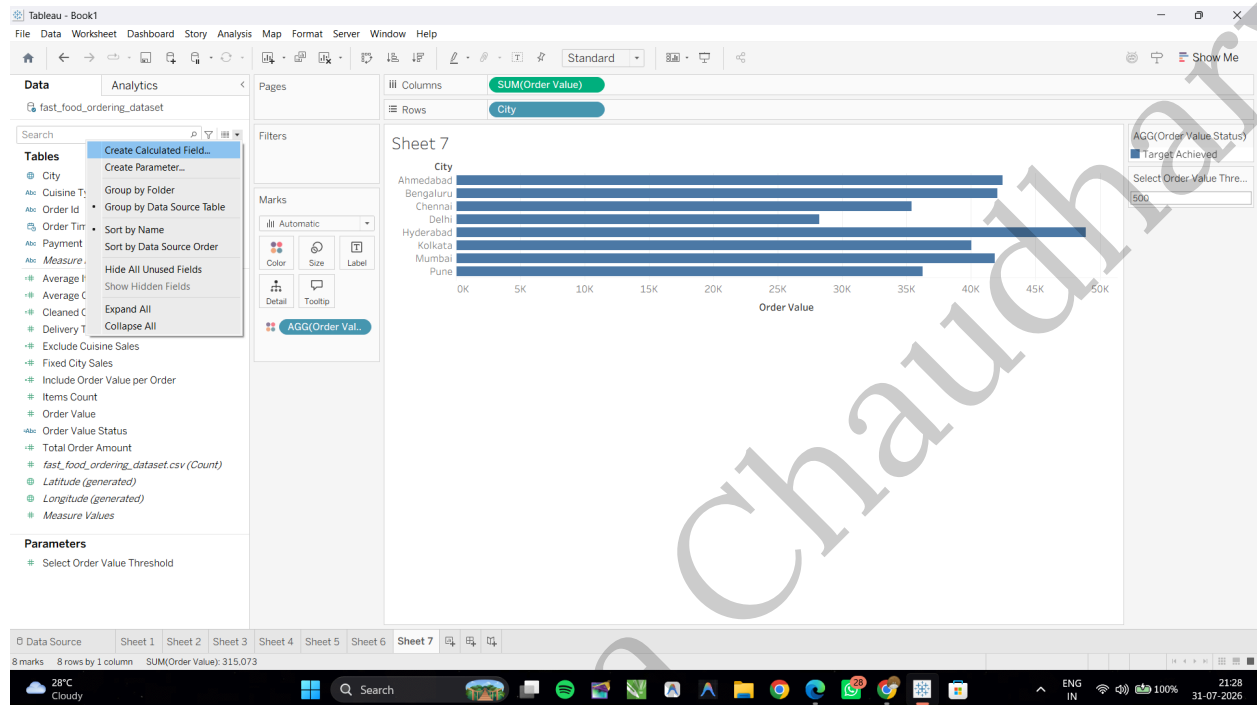


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Step 2: Standardizing Text Using UPPER()

Right-click the Data Pane, select Create Calculated Field, name it Standardized City, enter `UPPER([city])`, then click OK.

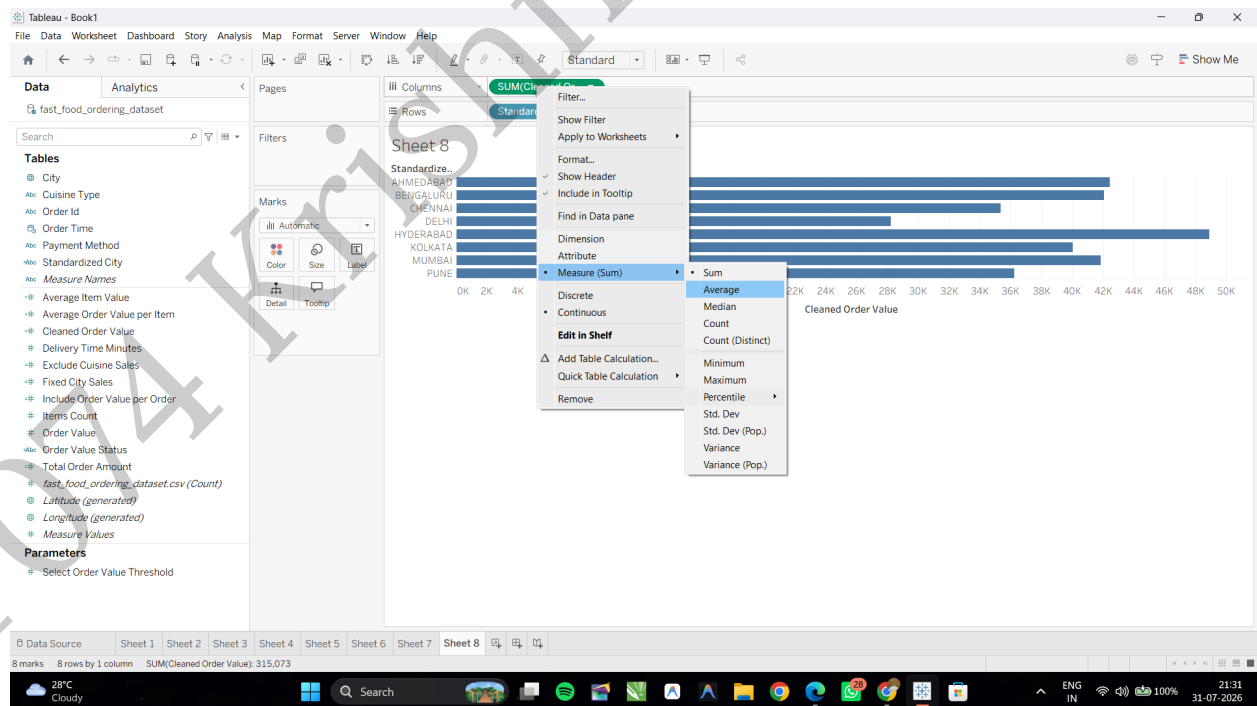
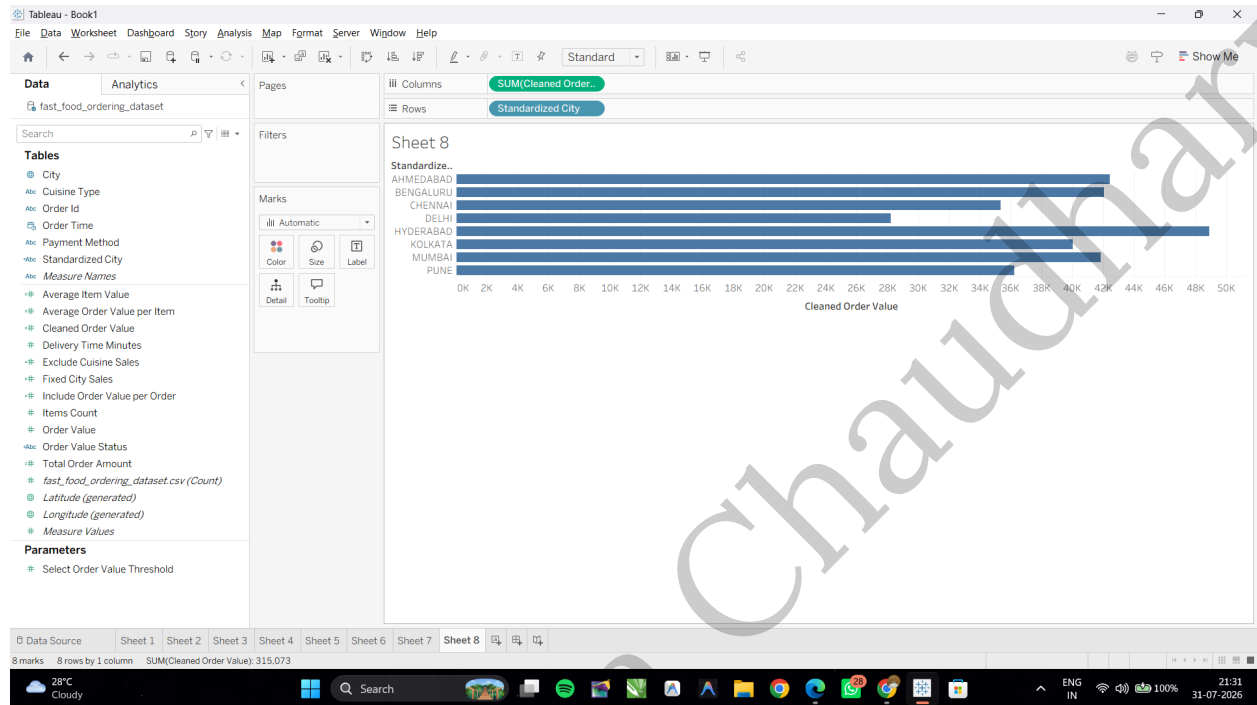


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Step 3: Putting the Cleaned Fields to Use

Open a new worksheet, drag Standardized City to the Rows shelf and Cleaned Order Value to the Columns shelf, then change SUM(Cleaned Order Value) to Average.



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